TETRAGONAL SHIMURA CURVES OVER Q AND GEOMETRICALLY: KNOWN FAMILIES AND EXPLICIT FINITE UNRESOLVED CANDIDATE LISTS

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Allowed standard prerequisites (editorial).

- Shimura curves, Hecke/Jacobian isogenies, gonality, Castelnuovo-Severi and CM points. Original source-local formulations, all referenced tables and complete bibliography retained.

1. Introduction

In [DLMS23], Dose–Lido–Mercuri–Stirpe provide an algorithm for computing the number of points on X over a finite field $\mathbb{F}_q$ for X a member of a wide family of modular curves and their quotients by Atkin–Lehner involutions. A main focus in the referenced article is on finding explicit curves which achieve the largest known point counts among curves of a fixed genus g over a finite field $\mathbb{F}_q$ of fixed cardinality. Indeed, the referenced authors found among the studied families of modular curves many new examples achieving the largest known point counts over small finite fields of characteristic up to 19 at the time of publication of their work based on data available at [vdGHLR09] (see [DLMS23, §6]). Their work is extended in [DLMS26] to a larger family including all the possible Atkin–Lehner quotients.

In the current work, we implement a similar algorithm for the Shimura curves $X_0^D(N)$ over $\mathbb{Q}$ and their Atkin–Lehner quotients, parameterizing abelian surfaces with level structure and with potential quaternionic multiplication by the indefinite quaternion algebra over $\mathbb{Q}$ of discriminant D. The case $D = 1$ recovers the modular curve setting of [DLMS23], so we take $D > 1$ in this work. Our main algorithm is achieved using an isogeny due to Ribet (see Theorem 2.3 below) in combination with the methods of [DLMS23], and it has the following form.

Algorithm 1.1.

Input: *A tuple (D, N, W, p, r), where:*

- *$D > 1$ is the discriminant of an indefinite quaternion algebra over $\mathbb{Q}$ (equivalently, a product of an even number of distinct rational primes);*
- *N is a positive integer coprime to D;*
- *W is a subgroup of the group $W_0(D, N) \leq \mathrm{Aut}(X_0^D(N))$ of Atkin–Lehner involutions on $X_0^D(N)$ (defined in Section 2.1), given by a list $[m_1, \ldots, m_n]$ of Hall divisors of DN so that $\{w_{m_1}, \ldots, w_{m_n}\}$ is a generating set for W;*
- *$p \nmid DN$ is a rational prime;*
- *r is a positive integer.*

Output: *The number $\#\left(X_0^D(N)/W\right)(\mathbb{F}_{p^r})$ of $\mathbb{F}_{p^r}$-rational points on the base change of the curve $X_0^D(N)/W$ to the finite field $\mathbb{F}_{p^r}$.*

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Remark 1.2. As Algorithm 1.1 is described above, it includes computations of eigenvalues for the action of Hecke operators on a suitable basis of cusp forms of weight 2 associated to isogeny factors of the Jacobian of the modular curve $X_0(DN)$. In practice, for the purpose of speed, one can take such data as input. For computations described in this paper, we use such data obtained from [LMF25] (see Section 4 for further discussion).

Among the family of curves studied, we find 116 curves achieving, to our knowledge and based on current information on [vdGHLR09], record-breaking point counts beyond those observed in [DLMS23] and [DLMS26] (see Table 3). Additionally, we find 898 curves that are maximal (i.e., they attain the greatest possible point count) for given pairs (genus, finite field). In these cases, there were already known curves attaining these number of points but it was not previously known that the counts were realized by Atkin–Lehner quotients of Shimura curves. Furthermore, we find (at least) 4 previously unknown isomorphism classes of maximal curves.

Our implementation of Algorithm 1.1 is also strongly motivated by potential applications to arithmetic properties of the curves $X_0^D(N)$ and their quotients. An important detail about this algorithm is that it does not require defining equations for the studied curves, as is true for the point count algorithm for modular curves of [DLMS23] and [DLMS26]. Our knowledge of explicit equations for Shimura curves in the case $D > 1$ is quite limited compared to that for modular curves, due in part to the absence of cuspidal points on these Shimura curves.

As a first example, we consider the automorphism groups of these curves over $\mathbb{Q}$. Restricting to squarefree level N, it is expected that $\mathrm{Aut}(X_0^D(N))$ consists only of Atkin–Lehner involutions for all but finitely many pairs (D, N). A result of González [Gon17] (Lemma 5.1 below) provides a restriction on the automorphism group of a curve over $\mathbb{Q}$ based on its point counts over finite fields, and using this restriction and other restrictions from work of Kontogeorgis–Rotger [KR08] (Lemma 2.7 below) we implement algorithmic checks of sufficient conditions for $X_0^D(N)$ and its quotients to have no non-Atkin–Lehner involutions. We reach the following computational result (which appears in Section 5 as Corollary 5.4).

Theorem 1.3. *Let $D > 1$ be the discriminant of an indefinite quaternion algebra over $\mathbb{Q}$, let N be a positive integer coprime to D such that*

- *$X_0^D(N)$ has genus $g > 2$ and*
- *$DN \leq 10000$,*

and let W be a subgroup of Atkin–Lehner involutions of $\mathrm{Aut}(X_0^D(N))$.

(1) If (D, N) does not appear among the 12 pairs in Table 7, then we have that $\mathrm{Aut}(X_0^D(N)) = W_0(D, N)$.

(2) If (D, N) does not appear among the 1321 pairs in Table 8, then $\mathrm{Aut}(X_0^D(N)/W)$ consists only of Atkin–Lehner involutions. Hence

$$\mathrm{Aut}(X_0^D(N)/W) \cong (\mathbb{Z}/2\mathbb{Z})^{\omega(DN)-\mathrm{ord}_2(|W|)},$$

where $\omega(n)$ denotes the number of distinct prime divisors of an integer n.

Remark 1.4. There is a total of 10609 coprime pairs (D, N) with $D > 1$ satisfying the bulleted items in Theorem 1.3.

For a curve over $\mathbb{Q}$, point counts over finite fields also supply lower bounds on gonalities over $\mathbb{Q}$ (see Lemma 3.3). Explicit point count computations provide improvements in many cases over known bounds, such as Abramovich's bound in terms of the genus [Abr96] (Theorem 6.15 below), while other techniques from the modular curve setting (see, for example, [DvH14]) do not transfer over due to lack of cuspidal points.

Gonality bounds and computations are important in applications to low-degree points on modular and Shimura curves, and we expect Algorithm 1.1 to be integral in advancing our knowledge in this area. Shimura curves lack real points [Shi75, Theorem 0] and hence lack points of any odd degree over $\mathbb{Q}$. The curves $X_0^D(N)$ with $D > 1$ with infinitely many degree 2 points were determined by Padurariu–Saia [PS25a, Theorem 6.3], building on work of Ogg [Ogg83], Guo–Yang [GY17], and Rotger [Rot02]. To determine those curves with infinitely many degree 4 points, a natural first question is to ask which curves $X_0^D(N)$ have gonality 4 over $\mathbb{Q}$, that is, which admit a degree 4 map to the projective line over $\mathbb{Q}$. Using our point counts in combination with other methods, we come close to this determination.

Theorem 1.5. *Let $D > 1$ be the discriminant of an indefinite quaternion algebra over $\mathbb{Q}$ and let N be a positive integer which is relatively prime to D.*

(1) If the Shimura curve $X_0^D(N)$ is geometrically tetragonal and it is not among the 161 geometrically tetragonal curves $X_0^D(N)$ with (D, N) listed in Table 9, Table 10, and Table 11, then (D, N) must be among the 29 pairs listed in Table 1.

(2) If the Shimura curve $X_0^D(N)$ is tetragonal over $\mathbb{Q}$ and it is not among the 141 tetragonal curves $X_0^D(N)$ with (D, N) listed in Part (2) of Proposition 6.17 and in Table 12, then (D, N) must be among the 32 pairs listed in Table 2.

This work is organized as follows. In Section 2, we provide useful background and known results on Shimura curves and their automorphism groups. We do the same regarding gonalities of curves in Section 3. We then outline Algorithm 1.1 in Section 4. The remaining sections are devoted to applications: Section 5 and Section 6 feature our applications to automorphism groups of Shimura curves and tetragonal Shimura curves, respectively, and we study curves achieving largest known point counts in Section 7.

All computations described in this paper were performed using the Magma computer algebra system [BCP97]. All related code can be found in the GitHub repository [MPSS26].

Acknowledgments

We thank John Voight for helpful discussion and for the initial suggestion of the project. We thank Valerio Dose and Guido Lido for their contribution to the code for the point counts over finite fields. P.M. is supported by the "National Group for Algebraic and Geometric Structures, and their Applications" (GNSAGA - INdAM). F.S. gratefully acknowledges support provided by an AMS-Simons Travel Grant.

2. Shimura curves

A *quaternion algebra* over a field F is a central simple algebra of dimension 4 over F. We call a quaternion algebra B over the field $\mathbb{Q}$ of rational numbers *indefinite* if it is split at the single real place of $\mathbb{Q}$, i.e., if

$$B \otimes_{\mathbb{Q}} \mathbb{R} \cong M_2(\mathbb{R}).$$

A quaternion algebra B over $\mathbb{Q}$ is determined up to isomorphism by its discriminant D, that is the product of the rational primes at which B is ramified. The indefinite quaternion discriminants over $\mathbb{Q}$ are exactly those integers which are products of an even number of distinct rational primes. The case $D = 1$, for example, is that of the split quaternion algebra $M_2(\mathbb{Q})$.

Let D be the discriminant of an indefinite quaternion algebra B over $\mathbb{Q}$, let N be a positive integer which is relatively prime to D, and let $\mathcal{O}_N$ be an Eichler order of level N

in B (as defined in [Voi21, §23.1.3]). Consider, over the algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$, the moduli problem of parameterizing pairs (A, ι), where:

- A is an abelian surface, and
- $\iota\colon \mathcal{O}_N \hookrightarrow \mathrm{End}(A)$ is an embedding.

A theorem of Shimura [Shi67, Main Theorem 1] states that this problem is represented by a scheme which admits a canonical model over $\mathbb{Q}$, and we let $X_0^D(N)$ denote this coarse moduli scheme over $\mathbb{Q}$. The case $D = 1$ of $X_0^1(N)$ recovers the modular curve setting, parameterizing elliptic curves with torsion level structure. When $D > 1$, these curves lack the rational cusps the modular curves $X_0(N)$ possess and, moreover, they have no real points [Shi75, Theorem 0]: if $D > 1$, then $X_0^D(N)(\mathbb{R}) = \varnothing$.

By the work of Drinfeld, for which we refer to [BC91, Buz97], there exists a flat proper model of $X_0^D(N)$ over $\mathbb{Z}$, extending the moduli interpretation, which is smooth over $\mathbb{Z}\left[\frac{1}{DN}\right]$. For this work, it is important to note that $X_0^D(N)$ has good reduction at primes $p \nmid DN$.

For N a positive integer, we have a natural modular covering map $X_0^D(N) \to X_0^D(1)$ with degree $\psi(N)$ given by the Dedekind Psi function:

$$\psi(N) = N \prod_{\substack{p \mid N \\ \text{prime}}} \left(1 + \frac{1}{p}\right).$$

Moreover, these natural maps are compatible: for $N_1 \mid N_2$ there is a natural map $X_0^D(N_2) \to X_0^D(N_1)$ of degree $\psi(N_2)/\psi(N_1)$. We have the following formula for the genus of the curve $X_0^D(N)$ (see, for example, [Voi21, Thm. 39.4.20]).

Proposition 2.1. *For $D > 1$ the discriminant of an indefinite quaternion algebra over $\mathbb{Q}$, for N a positive integer coprime to D, and for $k \in \{3, 4\}$, define*

$$e_k(D, N) := \prod_{\substack{p \mid D \\ \text{prime}}} \left(1 - \left(\frac{-k}{p}\right)\right) \prod_{\substack{\ell \mid N \\ \text{prime}}} \left(1 + \left(\frac{-k}{\ell}\right)\right) \prod_{\substack{\ell^2 \mid N, \\ \ell \text{ prime}}} \left(\frac{-k}{\ell}\right).$$

The genus of $X_0^D(N)$ is given by

$$g(X_0^D(N)) = 1 + \frac{\phi(D)\psi(N)}{12} - \frac{e_4(D, N)}{4} - \frac{e_3(D, N)}{3},$$

where ϕ denotes the Euler totient function.

2.1. **Atkin–Lehner involutions and Ribet's isogeny.** For the entirety of this work, for a curve X over a field F, when we write $\mathrm{Aut}(X)$ or refer to the automorphisms of X we mean, unless otherwise specified, the group of automorphisms of X defined over the base field F.

The automorphism group $\mathrm{Aut}(X_0^D(N))$ of the curve $X_0^D(N)$ over $\mathbb{Q}$ has a subgroup consisting of involutions which are natural from the moduli-theoretic perspective. This subgroup is that of the *Atkin–Lehner involutions*, and it contains a non-trivial involution $w_m(D, N)$ for each divisor $m \parallel DN$ with $m \neq 1$, where $m \parallel DN$ means that m is a Hall divisor, i.e., that $m \mid DN$ and $\gcd(m, DN/m) = 1$. We denote this subgroup by $W_0(D, N) \leq \mathrm{Aut}(X_0^D(N))$. For more background on Atkin–Lehner involutions, we refer to [Ogg77] and [Ogg85, §2]; here we give a quick definition from the quaternion arithmetic vantage and state what we need regarding the structure of this group.

Realizing $X_0^D(N)$ as the coarse moduli scheme attached to the moduli problem of parameterizing abelian surfaces with a quaternionic multiplication structure by an Eichler

order $\mathcal{O}_N$ of level N in the quaternion algebra B of discriminant D, we have the following subgroup of the normalizer of $\mathcal{O}_N$

$$N_{B^\times_{>0}}(\mathcal{O}_N) := \{u \in B^\times : \mathrm{nrd}(u) > 0 \text{ and } u^{-1}\mathcal{O}_N u = \mathcal{O}_N\},$$

where nrd is the reduced norm, and we realize the group of Atkin–Lehner involutions as

$$W_0(D,N) := N_{B^\times_{>0}}(\mathcal{O}_N)/\mathbb{Q}^\times\mathcal{O}_N^\times \leq \mathrm{Aut}(X_0^D(N)).$$

Recalling that $\omega(n)$ denotes the number of distinct prime divisors of an integer n, this group is a finite abelian 2-group of the form

$$W_0(D,N) = \{w_m(D,N) : m \parallel DN\} \cong (\mathbb{Z}/2\mathbb{Z})^{\omega(DN)},$$

though we simply write w_m for $w_m(D,N)$ for brevity when the curve $X_0^D(N)$ whose automorphisms we are considering is clear from context.

For a subgroup $W \leq W_0(D,N)$, we call the quotient $X_0^D(N)/W$ an *Atkin–Lehner quotient* of $X_0^D(N)$. We also refer to the elements of the subgroup of $\mathrm{Aut}\left(X_0^D(N)/W\right)$ induced by the elements of $W_0(D,N)$ as Atkin–Lehner involutions.

Ogg determined the set of fixed points for each Atkin–Lehner involution on the curve $X_0^D(N)$: these are pairwise disjoint for distinct involutions and they consist of complex multiplication (CM) points by certain imaginary quadratic orders.

Proposition 2.2. [Ogg83, p. 283] *Let $m > 1$ with $m \parallel DN$. The fixed points of the Atkin–Lehner involution w_m acting on $X_0^D(N)$ are points with* CM *by the following imaginary quadratic orders:*

$$R = \begin{cases} \mathbb{Z}[i] \text{ and } \mathbb{Z}[\sqrt{-2}] & \text{if } m = 2, \\ \mathbb{Z}\left[\frac{1+\sqrt{-m}}{2}\right] \text{ and } \mathbb{Z}[\sqrt{-m}] & \text{if } m \equiv 3 \pmod 4, \\ \mathbb{Z}[\sqrt{-m}] & \text{otherwise.} \end{cases}$$

In the referenced work, Ogg provides the count for the number of fixed points of w_m with CM by R for each of the orders R listed for m in Proposition 2.2. Letting $\mathcal{O}_N$ denote a fixed Eichler order of level N in the indefinite quaternion algebra over $\mathbb{Q}$ of discriminant D, the count is given as

$$h(R) \prod_{\substack{p \mid \frac{DN}{m} \\ \text{prime}}} \nu_p(R, \mathcal{O}_N),$$

where $h(R)$ is the class number of the order R and $\nu_p(R,\mathcal{O}_N)$ is the number of inequivalent optimal embeddings (see [Voi21, Definition 30.3.2]) of the localization R_p of R at p into the localization $(\mathcal{O}_N)_p$ of $\mathcal{O}_N$ at p. The counts of these optimal embeddings are given in [Ogg83, Theorem 2].

Using Proposition 2.2 and the Riemann–Hurwitz theorem in combination with the genus formula for $X_0^D(N)$ (Proposition 2.1), one can compute the genus of every Atkin–Lehner quotient of $X_0^D(N)$. A function handling these genus computations is implemented in the file `quot_genus.m` in [MPSS26].

The following theorem of Ribet, relating the Jacobian of $X_0^D(N)$ to a sub-quotient of the Jacobian of the modular curve $X_0(DN)$, is crucial to our implementation of a finite field point count algorithm for $X_0^D(N)$ in Section 4. While Ribet's original result in [Rib80], as well as the work of Bertolini–Darmon [BD96, §1] in which our statements regarding the Atkin–Lehner action appear, each contain a squarefree restriction on the level N, this hypothesis can be removed. One can find the relevant results on bases of

spaces of modular forms and embeddings of quaternion orders in the work of Hijikata–Pizer–Shemanske [HPS89b, HPS89a], and in particular we refer the reader to the relevant statement reproved by Martin in [Mar20, Theorem 1.1 (ii)]. One can also see [Rob89, Corollary 3.3] for an alternate proof of the general version.

Theorem 2.3. [Rib80, BD96, HPS89b, HPS89a] *Let $w_m(1, DN)$ denote the involution on $X_0(DN)$ and let $w_m(D, N)$ denote the involution on $X_0^D(N)$ corresponding to a Hall divisor m of DN. There is an isogeny defined over $\mathbb{Q}$*

$$\Psi\colon \operatorname{Jac}(X_0(DN))^{D-\text{new}} \longrightarrow \operatorname{Jac}(X_0^D(N))$$

such that the following holds: for each $m \parallel DN$ and each isogeny factor A_f of $\operatorname{Jac}(X_0^D(N))$, letting $\epsilon_f(D, N)$ and $\epsilon_f(1, DN)$ denote, respectively, the signs of the actions of $w_m(D, N)$ and $w_m(1, DN)$ on A_f, we have

$$\epsilon_f(D, N) = (-1)^{\omega(\gcd(D,m))}\epsilon_f(1, DN). \tag{1}$$

Remark 2.4. The Jacobian $\operatorname{Jac}(X_0(DN))$ of $X_0(DN)$ decomposes into irreducible isogeny factors A_f, which correspond to distinguished cusp forms f. To obtain the D-new part, we take those factors with conductor divisible by D. Since A_f, when f is a newform of level DN, has conductor $(DN)^{\dim A_f}$ (see [KW09a] and [KW09b] for the proof of Serre's modularity conjecture, and [Ser87, Théorème 5] for its consequence we are interested in), hence, by the classical theory of modular curves and oldforms (see [AL70, Theorem 5] and [DS05, Chapters 5 and 6]), we have

$$\operatorname{Jac}(X_0(DN))^{D-\text{new}} \sim \prod_{n|N} \operatorname{Jac}^{\text{new}}(X_0(Dn))^{\sigma_0\left(\frac{N}{n}\right)},$$

where $\sigma_0(m)$ is the sum of positive divisors of the integer m and $\operatorname{Jac}^{\text{new}}(X_0(Dn))$ denotes the new part of $\operatorname{Jac}(X_0(Dn))$, i.e., the part where the isogeny factors A_f are associated to newforms f of level Dn.

The isogeny factors of the Jacobian of an Atkin–Lehner quotient $X_0^D(N)/W$, and the corresponding cusp forms, can then be determined by the action of W on these isogeny factors as given in Equation (1).

2.2. **Known results on automorphism groups.** It is expected that

$$\operatorname{Aut}(X_0^D(N)) = W_0(D, N) \tag{2}$$

for all but finitely many Shimura curves $X_0^D(N)$ with squarefree level N (see [KR08, Conjecture 1.1]). This is known if one restricts to the case $D = 1$ of the modular curves $X_0(N)$, even for general N, by work of Ogg [Ogg77], Kenku–Momose [KM88], Elkies [Elk90], and Harrison [Har11]. Here, the only curves $X_0(N)$ of genus at least 2 with automorphisms which are not Atkin–Lehner involutions are those of level $N \in \{37, 63, 108\}$.

In Section 5, we provide methods for checking this equality, along with related experimental results for $D > 1$. Here, we recall related results of Kontogeorgis–Rotger.

Proposition 2.5. [KR08, Proposition 1.5] *Suppose N is squarefree and coprime to D and let $W \leq W_0(D, N)$. If $g(X_0^D(N)/W) \geq 2$, then all automorphisms of $X_0^D(N)/W$ over $\overline{\mathbb{Q}}$ are defined over $\mathbb{Q}$ and*

$$\operatorname{Aut}(X_0^D(N)/W) \cong (\mathbb{Z}/2\mathbb{Z})^s$$

for some $s \geq \omega(DN) - \operatorname{ord}_2(|W|)$.

Proposition 2.6. [KR08, Corollary 1.8] *Suppose N is squarefree and coprime to D. If $g(X_0^D(N)) \geq 2$, then*

$$\mathrm{Aut}(X_0^D(N)) \cong (\mathbb{Z}/2\mathbb{Z})^s$$

for some $s \in \{\omega(DN), \omega(DN)+1\}$.

In the following lemma, giving sufficient criteria for the equality $\mathrm{Aut}(X_0^D(N)) = W_0(D,N)$, the numbers $e_k(D,N)$ for $k \in \{3,4\}$ are as in Proposition 2.1.

Lemma 2.7. [KR08, Theorems 1.6 and 1.7] *Suppose that N is squarefree and coprime to D and that $g := g(X_0^D(N)) \geq 2$. If at least one of the following statements holds:*

(1) $e_3(D,N) = e_4(D,N) = 0$;
(2) $3 \mid DN$, for all primes $p \mid N$ we have $\left(\frac{-3}{p}\right) \neq -1$, and for at most one prime $\ell \mid D$ we have $\left(\frac{-3}{\ell}\right) = 1$;
(3) $\omega(DN) = \mathrm{ord}_2(g-1) + 2$;
(4) there is a prime $p \nmid 2DN$ such that $\omega(DN) = \mathrm{ord}_2\left(|X_0^D(N)(\mathbb{F}_p)|\right) + 1$;

then $\mathrm{Aut}(X_0^D(N)) = W_0(D,N)$.

3. Gonalities

3.1. Basic properties.

Definition 3.1. Let F be a perfect field with algebraic closure $\overline{F}$, and let X be a curve over F. The *gonality* $\gamma_F(X)$ of X (over F) is the least degree of a map from X to the projective line $\mathbb{P}^1_F$ over F. A map $X \to \mathbb{P}^1_F$ over F of degree $\gamma_F(X)$ is called a *gonal map* for X.

For L/F a field extension, we have the base change of X to L

$$X_L := X \otimes_{\mathrm{Spec}\, K} \mathrm{Spec}\, L,$$

and we denote the gonality of this base change by

$$\gamma_L(X) := \gamma_L(X_L).$$

In particular, the *geometric gonality* of X is defined to be $\gamma_{\overline{F}}(X)$. We recall the following general facts about gonalities which we use often.

Proposition 3.2. [Poo07, Proposition A.1] *Let X be a curve of genus g over a field F.*

(1) If L/F is a field extension, then $\gamma_L(X) \leq \gamma_F(X)$.
(2) If F is algebraically closed and L/F is a field extension, then $\gamma_L(X) = \gamma_F(X)$.
(3) If $g > 1$, then $\gamma_F(X) \leq 2g-2$. For each $g > 1$, there exist F and X/F for which equality holds.
(4) If $X(F) \neq \varnothing$, then $\gamma_F(X) \leq g+1$. If further $g \geq 2$, then $\gamma_F(X) \leq g$ (and, again, equality is possible for each g).
(5) If F is algebraically closed, then $\gamma_F(X) \leq \lfloor \frac{g+3}{2} \rfloor$. Equality holds for a general curve of genus g over F.
(6) If $\pi\colon X \to Y$ is a dominant rational map of curves over F, then

$$\gamma_F(Y) \leq \gamma_F(X) \leq \deg(\pi) \cdot \gamma_F(Y).$$

3.2. **Gonalities and reduction.** Let X be a curve over $\mathbb{Q}$, let p be a prime of good reduction for X, and let $q = p^r$ where r is a positive integer. We let $\gamma_{\mathbb{F}_q}(X)$ denote the gonality of the base change of X to $\mathbb{F}_q$. In this setting, we have the inequality

$$\gamma_{\mathbb{F}_q}(X) \leq \gamma_{\mathbb{Q}}(X),$$

as from a rational function on X over $\mathbb{Q}$ corresponding to a gonal map one obtains via base change a rational function over $\mathbb{F}_q$ of degree at most $\gamma_{\mathbb{Q}}(X)$. In particular, the Shimura curve $X_0^D(N)$ has good reduction at primes $p \nmid DN$, and, hence, if $q = p^r$, we get

$$\gamma_{\mathbb{F}_q}(X_0^D(N)) \leq \gamma_{\mathbb{Q}}(X_0^D(N)).$$

If $\widetilde{X}$ is a curve over $\mathbb{F}_q$ with gonality d and f is a gonal map for $\widetilde{X}$, then there are at most d points of degree 1 in the fiber of f above each of the $q+1$ rational points on $\mathbb{P}^1_{\mathbb{F}_q}$. Therefore, we have the following gonality bounds via point counts.

Lemma 3.3. *Let X be a curve over $\mathbb{Q}$, let p be a prime of good reduction for X, and let $q = p^r$ for r a positive integer. We have*

$$\gamma_{\mathbb{Q}}(X) \geq \gamma_{\mathbb{F}_q}(X) \geq \frac{\#X(\mathbb{F}_q)}{q+1}.$$

4. Finite fields point counts

In this section, we describe Algorithm 1.1, which is used for our computations of point counts. It is a variation of the algorithms contained in [DLMS23] and [DLMS26].

By Ribet's isogeny (Theorem 2.3 above), we can write the Jacobian of the Shimura curve $X_0^D(N)$ in terms of isogeny factors of the Jacobian of the modular curve $X_0(DN)$. In order to work with the latter object, we recall the following results from [DLMS23] and [DLMS26].

Lemma 4.1. *Let $\mathcal{F}$ be a finite subset of the set of the Galois orbits of normalized eigenforms of weight 2 of $\bigcup_{n>0} \mathcal{S}_2^{\mathrm{new}}(\Gamma_0(n))$, and let $J := \prod_{[f]\in\mathcal{F}} A_f^{m_f}$, where A_f is the abelian variety associated to Galois orbit $[f]$ of f (see* [DS05, Definition 6.6.3]*) and $m_f > 0$ being an integer. Moreover, let $V_\ell = \mathrm{Ta}_\ell(J) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ be the $\mathbb{Q}_\ell$-vector space associated to the Tate module of J for a prime ℓ. Then, the characteristic polynomial of Frob_p acting on V_ℓ is*

$$\prod_{[f]\in\mathcal{F}} \prod_{h\in[f]} (x^2 - a_p(h)x + p)^{m_f},$$

where $a_p(h)$ is the p-th Fourier coefficient of h.

Proof. See [DLMS23, Proposition 4.4]. □

To apply the previous lemma, we need to compute $a_p(h)$ and m_f. For the former, we use the data available in [LMF25]. For the latter, we use the following result.

Lemma 4.2. *Given a modular curve $X_0(N)$, let W be its group of Atkin–Lehner involutions. For each subgroup $K \leq W$, the Jacobian of the corresponding quotient satisfies the following isogeny relation over $\mathbb{Q}$*

$$\mathrm{Jac}(X_0(N)/K) \sim \prod_f A_f^{m_f},$$

where f ranges among the newforms of weight 2 and level d for divisors $d \mid N$ and the multiplicities m_f are as follows: let $K^\perp$ be the set of characters of W that are trivial on

K and, for each f, denote $v := \mathrm{ord}_\ell(N/d)$. Then

$$m_f := \sum_{\chi \in K^\perp} \prod_{\ell^e \| N} \left(\tfrac{v+1}{2} + \varepsilon_{f,\ell^e} \chi(w_{\ell^e}) \tfrac{1+(-1)^v}{4} \right),$$

with the indices ℓ^e varying among all prime powers exactly dividing N and ε_{f,ℓ^e} being the eigenvalue of the Atkin–Lehner involution w_{ℓ^e} applied to f.

Proof. This follows from [DLMS26, Theorem 2.2], taking $n_0 = N$ and $n_{\mathrm{ns}} = 1$. □

The steps of Algorithm 1.1 for counting the number of points of $X_0^D(N)/W$ over the finite field $\mathbb{F}_q$, with $q = p^k$ and $p \nmid DN$, are as follows.

(1) For every positive integer $n \mid N$, choose a basis $\mathcal{B}(nD)$ of newforms of level nD and let $\mathcal{B} := \bigcup_{n|N} \mathcal{B}(nD)$ be the union of these bases.
(2) For every $f \in \mathcal{B}$, compute the eigenvalues with respect to the Atkin–Lehner operators w_{ℓ^e}, for $\ell^e || DN$ (these data are available on the database [LMF25] for $DN \leq 10000$) and, then, use them for computing the multiplicity m_f as in Lemma 4.2. We point out that we need to switch ε_{f,ℓ^e} with $-\varepsilon_{f,\ell^e}$ if $\ell \mid D$ by Theorem 2.3.
(3) Compute the Hecke eigenvalue $a_p(f)$ for every $f \in \mathcal{B}$ (these data are also available on the database [LMF25] for $DN \leq 10000$).
(4) If $k = 1$, compute the Frobenius traces acting on A_f:

$$\mathrm{tr}(\mathrm{Frob}_p | A_f) = \sum_{h \in [f]} a_p(h),$$

where $[f]$ is the Galois orbit of f.
(5) If $k > 1$, for each $h \in [f]$, compute the (complex) roots $\alpha(h)$ and $\beta(h)$ of the polynomial $x^2 - a_p(h)x + p = 0$. Then compute

$$\mathrm{tr}(\mathrm{Frob}_q | A_f) = \sum_{h \in [f]} \alpha(h)^k + \beta(h)^k.$$

(6) Finally, compute

$$\#X_0^D(N)(\mathbb{F}_q) = q + 1 - \sum_{[f]} m_f \mathrm{tr}(\mathrm{Frob}_q | A_f),$$

where the sum is taken over the distinct Galois orbits for $f \in \mathcal{B}$ and m_f is defined in Lemma 4.2.

5. Automorphisms of Shimura curves

Let $X/\mathbb{Q}$ be a curve and let p be a prime of good reduction for X. The automorphism group of X over $\mathbb{Q}$ injects into the automorphism group of the reduction of X modulo p [Liu02, Proposition 3.38, Chapter 10]. In particular, if $p \nmid DN$, then the automorphism group of the curve $X_0^D(N)$ over $\mathbb{Q}$ injects into that of its reduction modulo p.

Point counts over the extensions $\mathbb{F}_q$ for q a power of p yield constraints on the automorphism group of a curve over $\mathbb{F}_p$. For instance, a constraint of this type, specific to the curves $X_0^D(N)$, is seen in Part (4) of Lemma 2.7. In this section, we use our point counts algorithm to implement checks on the automorphism group of $X_0^D(N)$ and compute evidence towards the conjecture mentioned in Section 2.2 (see Equation (2)).

By Proposition 2.5, we know that when N is squarefree and coprime to D the automorphism group of every Atkin–Lehner quotient of $X_0^D(N)$ consists of involutions. Of course, every Atkin–Lehner quotient of $X_0^D(N)$ has a non-trivial Atkin–Lehner involution *except possibly* for the full quotient $X_0^D(N)^* := X_0^D(N)/W_0(D,N)$. We expect that

this star quotient has trivial automorphism group for all but finitely many pairs (D, N), and the following instance of a result of González is a main tool we use to show this for a fixed pair.

Lemma 5.1. [Gon17, Equation (2.2)] *Let $q = p^r$ be a prime power, let X be a curve of genus $g \geq 2$ over the finite field $\mathbb{F}_q$, and suppose that X has a non-trivial automorphism of prime order ℓ over $\mathbb{F}_q$. For each integer $n \geq 1$, let*

$$A_n := X(\mathbb{F}_{q^n}) \setminus \bigcup_{i=1}^{n-1} X(\mathbb{F}_{q^i}).$$

Then

$$\sum_{n \geq 1, \gcd(n,\ell)=1} n \cdot \bmod(|A_n|/n, \ell) \leq \left\lfloor \frac{2g}{\ell - 1} \right\rfloor + 2,$$

where $\bmod(m, \ell)$ denotes the remainder upon division of m by ℓ.

To state our main computational results on automorphism groups, we set the following notation: let $\mathcal{S}$ denote the set of all pairs (D, N) consisting of an indefinite quaternion discriminant $D > 1$ over $\mathbb{Q}$ and a squarefree positive integer N coprime to D, such that

(1) $g(X_0^D(N)^*) > 2$ and
(2) $DN \leq 10000$.

The set $\mathcal{S}$ has cardinality $|\mathcal{S}| = 10609$.

Remark 5.2. The hypothesis (1) on the genus is necessary, as any curve $X_0^D(N)^*$ of genus $g \leq 2$ has a non-trivial automorphism, and this only excludes finitely many pairs (D, N) (see Section 6 for the relevant argument). Of course, we can only compute point counts for a finite list of curves, and the specific restriction (2) of $DN \leq 10000$ reflects the limits of the data we use of the eigenvalues of the Atkin–Lehner operators acting on newforms from the L-Functions and Modular Forms Database [LMF25], as discussed in Section 4.

Theorem 5.3. *The curve $X_0^D(N)^*$ has trivial automorphism group for all pairs $(D, N) \in \mathcal{S}$ except possibly for the 1321 pairs listed in Table 8.*

Proof. By Proposition 2.5, the only possible non-trivial automorphisms of $X_0^D(N)^*$ for $(D, N) \in \mathcal{S}$ are involutions. Lemma 5.1 shows that $\mathrm{Aut}(X_0^D(N)^*)$ is trivial for 9288 elements of $\mathcal{S}$. This is computed using our point counts algorithm, applying the lemma of González using prime powers $q = p^r$ with $p < 100$ a prime of good reduction for $X_0^D(N)$ and with powers r up to 100 when possible, and more generally as high as the pre-computed trace data we use allows. The code for these computations can be found in the file `aut_checks.m` in [MPSS26]. □

Corollary 5.4.

(1) Let $(D, N) \in \mathcal{S}$ be a pair which does not appear among the 1321 pairs in Table 8 and let $W \leq W_0(D, N)$. Then $\mathrm{Aut}(X_0^D(N)/W)$ consists only of Atkin–Lehner involutions, and hence

$$\mathrm{Aut}(X_0^D(N)/W) \cong (\mathbb{Z}/2\mathbb{Z})^{\omega(DN) - \mathrm{ord}_2(|W|)}.$$

(2) Let $(D, N) \in \mathcal{S}$ be a pair which does not appear among the 12 pairs in Table 7. Then $\mathrm{Aut}(X_0^D(N))$ consists only of Atkin–Lehner involutions, i.e.,

$$\mathrm{Aut}(X_0^D(N)) = W_0(D, N) \cong (\mathbb{Z}/2\mathbb{Z})^{\omega(DN)}.$$

Proof. By Proposition 2.5 we know that $\mathrm{Aut}(X_0^D(N)/W)$ is commutative. Therefore, if this group contains a non-Atkin–Lehner involution, then this involution would induce a non-trivial involution on the quotient $X_0^D(N)^*$. If (D, N) is a pair in Table 8, then this would contradict Theorem 5.3, giving Part (1). Part (2) then follows quickly from Part (1) by supplementing it with applications of Lemma 2.7 to the curves $X_0^D(N)$ for pairs (D, N) in Table 8. Code for these computations is again found in `aut_checks.m` in [MPSS26]. □

Remark 5.5. The computations performed for the proofs of Theorem 5.3 and Corollary 5.4 took roughly 1.66 hours of computational time on a 2020 m1 MacBook Pro. Access to further trace data for computing finite field point counts would allow for a similar result for curves up to a larger level bound, and would possibly allow us to improve our result in the level range $DN \leq 10000$ as well by allowing for computations of point counts over finite fields of larger cardinality.

We also note that the squarefree restriction on N put on pairs in $\mathcal{S}$ is *not* necessary in the use of our point count algorithm and Lemma 5.1. Rather, it is only needed in the use of Lemma 2.7 and in the transition from Theorem 5.3 to Corollary 5.4 using Proposition 2.5. We use these individual results more generally in Section 6, and functions applying each are available in the file `aut_checks.m` in [MPSS26]. We restrict to the set $\mathcal{S}$ for the above results as a sample of the power of our point counts in determining automorphism groups.

6. Tetragonal Shimura curves

Lemma 3.3 provides a way to obtain lower bounds on the gonality $\gamma_{\mathbb{Q}}(X_0^D(N))$ of $X_0^D(N)$ via point counts over finite fields. In this section, we apply this method in combination with other techniques to work towards determining all of the tetragonal and geometrically tetragonal Shimura curves $X_0^D(N)$. We begin by defining these notions and similarly making clear our convention for the terms hyperelliptic, bielliptic, tetragonal, and sub-hyperelliptic.

Definition 6.1. Let F be a perfect field and let X be a curve over F of genus $g(X) \geq 2$. We call X

- *hyperelliptic* if there exists a degree 2 map from X to $\mathbb{P}^1_F$ over F,
- *geometrically hyperelliptic* if its base change to $\overline{F}$ is hyperelliptic,
- *bielliptic* if there exists a degree 2 map from X to an elliptic curve over F,
- *geometrically bielliptic* if its base change to $\overline{F}$ is bielliptic,
- *tetragonal* if there exists a degree 4 map from X to $\mathbb{P}^1_F$ over F, and
- *geometrically tetragonal* if its base change to $\overline{F}$ is tetragonal.

Remark 6.2. Note that our definition of tetragonal simply means that the curve admits a degree 4 map to the projective line; it does not mean that the curve has gonality *exactly* 4, though it implies that it has gonality $2, 3$, or 4.

Definition 6.3. Let F be a perfect field. We call a curve X over F *sub-hyperelliptic* if it is hyperelliptic or has genus 0 or 1. We call X *geometrically sub-hyperelliptic* if its base change to $\overline{F}$ is sub-hyperelliptic.

We recall in the next two subsections the Castelnuovo–Severi criterion and some quick consequences as well as prior work on curves $X_0^D(N)$ of low gonality. These provide methods to narrow lists of candidate curves $X_0^D(N)$ prior to our application of point counts to a select finite list of candidates in the final subsection.

6.1. **The Castelnuovo–Severi inequality.** The Castelnuovo–Severi inequality is one of our main tools in proving certain curves $X_0^D(N)$ are not tetragonal. A proof of the version below, working over a perfect field, can be found in [KS24, Theorem 14].

Theorem 6.4 (Castelnuovo–Severi)**.** *Let F be a perfect field and let X, Y, and Z be curves over F. Let*

$$\pi_Y \colon X \to Y \qquad \text{and} \qquad \pi_Z \colon X \to Z$$

be non-constant morphisms defined over F. Then either

$$g(X) \leq \deg(\pi_Y) \cdot g(Y) + \deg(\pi_Z) \cdot g(Z) + (\deg(\pi_Y) - 1)(\deg(\pi_Z) - 1),$$

or there exists a curve X' over F and a morphism $X \to X'$ over F of degree greater than 1 through which both π_Y and π_Z factor.

We next collect applications of Theorem 6.4 that we will use in what follows.

Lemma 6.5. *Let X be a geometrically hyperelliptic curve over a perfect field F and let $f_1 \colon X \to C_1$ and $f_2 \colon X \to C_2$ be degree 2 maps over F to conics C_1 and C_2. Then there exists an isomorphism $f \colon C_1 \to C_2$ over F with $f_2 = f \circ f_1$.*

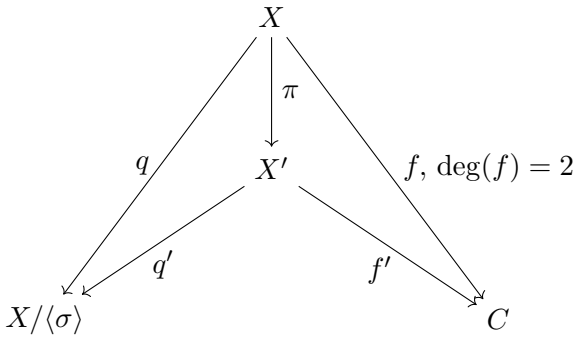
Proof. Note that the hypothesis that X is geometrically hyperelliptic implies that X has genus $g \geq 2$. Theorem 6.4 then implies that f_1 and f_2 factor through a common degree 2 map $h \colon X \to X'$ over F. We then have isomorphisms $f_1' \colon X' \to C_1$ and $f_2' \colon X' \to C_2$ over F so that $f_i = f_i' \circ h$. The claim then follows by letting $f = f_2' \circ (f_1')^{-1}$. □

Lemma 6.6. *Let X be a curve of genus g over a perfect field F possessing an involution σ, and suppose that the genus g_σ of the quotient $X/\langle\sigma\rangle$ satisfies $g_\sigma > 0$. If*

$$g > 2g_\sigma + 1,$$

then X is not geometrically hyperelliptic.

Proof. Let $q \colon X \to X/\langle\sigma\rangle$ denote the natural quotient map. Suppose to the contrary that there is a degree 2 map $f \colon X \to C$ to a conic C. Because of the assumed strict inequality, Theorem 6.4 tells us that we must have a commutative diagram of the following shape over F



with $\deg(\pi) > 1$. Hence $\deg(\pi) = 2$ and q' and f' are both isomorphisms, which implies that

$$g\left(X/\langle\sigma\rangle\right) = g(C) = 0$$

and contradicts our hypothesis. □

Lemma 6.7. *Let X be a geometrically bielliptic curve of genus $g \geq 6$ over a perfect field F and let $f_1 \colon X \to C_1$ and $f_2 \colon X \to C_2$ be degree 2 maps over F to genus 1 curves C_1 and C_2. Then there exists an isomorphism $f \colon C_1 \to C_2$ over F with $f_2 = f \circ f_1$.*

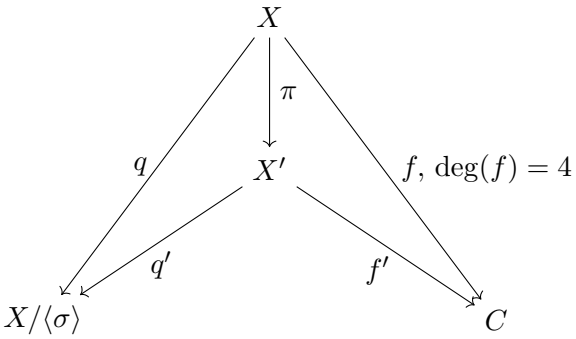
Proof. By the genus assumption on X, Theorem 6.4 then implies that f_1 and f_2 factor through a common degree 2 map $h\colon X \to X'$ over F. The proof then proceeds just as in Lemma 6.5. □

Lemma 6.8. *Let X be a geometrically tetragonal curve of genus g over a perfect field F with a geometrically tetragonal map $f\colon X \to C$ over F to a conic C. Let σ be an involution on X, and let g_σ denote the genus of $X/\langle\sigma\rangle$. If*

$$g > 2g_\sigma + 3, \tag{3}$$

then $X/\langle\sigma\rangle$ is geometrically sub-hyperelliptic.

Proof. Let $q\colon X \to X/\langle\sigma\rangle$ denote the natural quotient map. By Theorem 6.4, the assumed inequality implies that we have a commutative diagram over F of the following form:



with $\deg(\pi) = 2$. Here, q' must be an isomorphism, and thus

$$f' \circ (q')^{-1}\colon X/\langle\sigma\rangle \longrightarrow C$$

is a degree 2 map, proving the claim. □

Lemma 6.9. *Let X be a geometrically tetragonal curve of genus g over a perfect field F, with geometrically tetragonal maps $f_1\colon X \to C_1$ and $f_2\colon X \to C_2$ to conics C_1 and C_2 over F. If $g \geq 10$, then either*

- *X is geometrically hyperelliptic,*
- *X is geometrically bielliptic, or*
- *there exists an isomorphism $f\colon C_1 \to C_2$ over F with $f_2 = f \circ f_1$.*

Proof. By Theorem 6.4, the maps f_1 and f_2 are either equal up to composition with an isomorphism between C_1 and C_2 over F or they factor over F through a common degree 2 map $h\colon X \to X'$. In the latter case, the curve X' is evidently geometrically sub-hyperelliptic over F. In particular, if X' has genus 0, then X is geometrically hyperelliptic, and if X' has genus 1, then X is geometrically bielliptic.

The only remaining possibility is that X' is geometrically hyperelliptic with genus $g' \geq 2$. In this case, applying Lemma 6.5, we get that f_1 and f_2 are equal up to composition with an isomorphism $C_1 \to C_2$ over F. □

Corollary 6.10. *Let X be a geometrically tetragonal curve of genus $g \geq 10$ over a perfect field F. Assume that X is neither geometrically hyperelliptic nor geometrically bielliptic and that there exists a degree 4 map $f\colon X \to C$ over F to a conic C with $C(F) = \varnothing$. Then X is not tetragonal over F.*

Proof. By Lemma 6.9, we know with our hypotheses that if $h\colon X \to C'$ is another degree 4 map over F to a conic C', then C and C' are isomorphic over F. Therefore, C' is not isomorphic to $\mathbb{P}^1_F$. □

The next result is useful in the application of Corollary 6.10.

Lemma 6.11. *Let X be a curve over $\mathbb{Q}$ and let $q = p^r$ a power of a prime p of good reduction for X.*

(1) If X is geometrically hyperelliptic, then

$$X(\mathbb{F}_q) \leq 2q + 2.$$

(2) If X is geometrically bielliptic of genus $g \geq 6$, then

$$X(\mathbb{F}_q) \leq 2q + 2 + 4\sqrt{q}.$$

Proof.

(1) If the statement said "hyperelliptic over $\mathbb{Q}$" rather than "geometrically hyperelliptic," then it would follow immediately from Lemma 3.3. The point here is that the uniqueness of a geometrically hyperelliptic involution (as follows from Lemma 6.5) makes the result true as stated. The geometrically hyperelliptic involution on X must be defined over $\mathbb{Q}$ by uniqueness and, hence, X has a degree 2 map to a conic C over $\mathbb{Q}$ (which may have no rational points). Therefore, we get a map from $X_{\mathbb{F}_q}$ to a conic $\widetilde{C}$ over $\mathbb{F}_q$, as the automorphism group of X over $\mathbb{Q}$ injects into that of the reduction modulo p [Liu02, Proposition 3.38, Chapter 10]. We know that $\widetilde{C} \cong \mathbb{P}^1_{\mathbb{F}_q}$, as $\widetilde{C}$ is a smooth conic over a finite field, so $X_{\mathbb{F}_q}$ is hyperelliptic over $\mathbb{F}_q$ and the argument from Lemma 3.3 applies.

(2) This part is similar. Here, our assumption on the genus provides that the geometrically bielliptic involution on X is unique by Lemma 6.7. We then have a degree 2 map $X \to C$ over $\mathbb{Q}$ where C is a smooth genus 1 curve. This in turn gives a degree 2 map over $\mathbb{F}_q$ from $X_{\mathbb{F}_q}$ to a smooth genus 1 curve $\widetilde{C}$. By the Hasse bound, $\widetilde{C}$ has at most $q + 1 + 2\sqrt{q}$ over $\mathbb{F}_q$, and there are at most 2 $\mathbb{F}_q$-points in the fiber above each under the degree 2 map from $X_{\mathbb{F}_q}$.

□

The following consequence of the Riemann–Hurwitz theorem is helpful in the application of Lemma 6.11.

Proposition 6.12. [Has97, Proposition 4] *Let $X/\mathbb{C}$ be a hyperelliptic curve of genus g. Let σ be an involution on X and let g_σ be the genus of $X/\langle\sigma\rangle$. Suppose $g_\sigma > 0$. If g is even, then $g_\sigma = g/2$. If g is odd, then $g_\sigma \in \left\{\frac{g-1}{2}, \frac{g+1}{2}\right\}$.*

Finally, the following result of Najman–Orlić, which is a corollary of the Tower Theorem (see [NO24, Theorem 4.4]), will be useful in proving that a few curves are not geometrically tetragonal.

Proposition 6.13. [NO24, Corollary 4.6]

(1) Let C be a curve over $\mathbb{Q}$ of genus $g \geq 5$ which is geometrically trigonal and such that $C(\mathbb{Q}) \neq \emptyset$. Then C is trigonal over $\mathbb{Q}$.

(2) Let C be a curve over $\mathbb{Q}$ of genus $g \geq 10$ with $\gamma_{\overline{\mathbb{Q}}}(X) = 4$ and such that $C(\mathbb{Q}) \neq \emptyset$. Then $\gamma_{\mathbb{Q}}(X) = 4$.

6.2. **Known results on Shimura curves of low gonality.** The complete list of curves $X_0^D(N)$ with $D > 1$ and $\gcd(D, N) = 1$ of gonality $\gamma_{\mathbb{Q}}(X_0^D(N)) \leq 2$ and genus at least 2 is known. Equivalently, all such curves satisfying $\gamma_{\mathbb{Q}}(X_0^D(N)) \leq 3$ are known, since $\mathbb{Q}$-gonality of these curves is always even as $X_0^D(N)(\mathbb{R}) = \varnothing$ for $D > 1$. So too are those of geometric gonality $\gamma_{\overline{\mathbb{Q}}}(X_0^D(N)) \leq 3$. We collect these results in the following proposition.

Proposition 6.14.

(1) We have $\gamma_{\overline{\mathbb{Q}}}(X_0^D(N)) = 1$ *(i.e.,* $g(X_0^D(N)) = 0$*) if and only if*

$$(D, N) \in \{(6, 1), (10, 1), (22, 1)\}.$$

(2) We have $g(X_0^D(N)) = 1$ *if and only if* (D, N) *belongs to*

$$\{(6, 5), (6, 7), (6, 13), (10, 3), (10, 7), (14, 1), (15, 1), (21, 1), (33, 1), (34, 1), (46, 1)\}.$$

(3) We have $g(X_0^D(N)) \geq 2$ *and* $\gamma_{\mathbb{Q}}(X_0^D(N)) = 2$ *if and only if* (D, N) *belongs to*

$$\{(6, 11), (6, 19), (6, 29), (6, 31), (6, 37), (10, 11), (10, 23), (14, 5), (15, 2), (22, 3), (22, 5),$$
$$(26, 1), (35, 1), (38, 1), (39, 1), (39, 2), (51, 1), (55, 1), (58, 1), (62, 1), (69, 1), (74, 1), (86, 1),$$
$$(87, 1), (94, 1), (95, 1), (111, 1), (119, 1), (134, 1), (146, 1), (159, 1), (194, 1), (206, 1)\}.$$

(4) We have $g(X_0^D(N)) \geq 2$ *and* $\gamma_{\overline{\mathbb{Q}}}(X_0^D(N)) = 2 < \gamma_{\mathbb{Q}}$ *if and only if*

$$(D, N) \in \{(6, 17), (10, 13), (10, 19), (14, 3), (15, 4), (21, 2), (26, 3), (57, 1), (82, 1), (93, 1)\}.$$

(5) We have $\gamma_{\overline{\mathbb{Q}}}(X_0^D(N)) = 3$ *if and only if*

$$(D, N) \in \{(106, 1), (118, 1)\}.$$

Proof. Parts (1) and (2) follow from standard genus bounds and computations. Parts (3) and (4) are work of Ogg [Ogg83] and Guo–Yang [GY17]. Part (5) follows from [PS25a, Theorem 7.4] and the prior parts. □

6.3. **Proof of Theorem 1.5.** A main tool for us is the following lower bound on the geometric gonality of $X_0^D(N)$. This is a special case of a more general theorem of Abramovich applying to all Shimura curves, where we have improved the constant appearing using the best known result on Selberg's eigenvalue conjecture [Kim03, p. 176].

Theorem 6.15. [Abr96, Theorem 1.1] *For the Shimura curve* $X_0^D(N)$*, we have*

$$\gamma_{\mathbb{C}}(X_0^D(N)) \geq \frac{975}{8192}\big(g(X_0^D(N)) - 1\big).$$

If $X_0^D(N)$ is geometrically tetragonal, then

$$\gamma_{\mathbb{C}}(X_0^D(N)) \leq 4.$$

It follows from Theorem 6.15 that $g(X_0^D(N)) \leq 34$. We have the following lower bound on this genus in terms of the product DN.

Lemma 6.16. [Sai24, Lemma 10.6] *Let* γ *denote the Euler–Mascheroni constant. For* $D > 1$ *the discriminant of an indefinite quaternion algebra over* $\mathbb{Q}$ *and* N *a positive integer coprime to* D*, we have*

$$g(X_0^D(N)) > 1 + \frac{DN}{12}\left(\frac{1}{e^{\gamma}\log\log(DN) + \frac{3}{\log\log 6}}\right) - \frac{7\sqrt{DN}}{3}.$$

From Lemma 6.16, we find that if $DN > 77416$, then $g(X_0^D(N)) > 34$ and, hence, $\gamma_{\mathbb{Q}}(X_0^D(N)) > 4$, so $X_0^D(N)$ is not tetragonal over $\mathbb{Q}$. For each pair (D, N), with $D > 1$ an indefinite quaternion discriminant and N a positive integer coprime to D, satisfying $DN \leq 77416$, we check the inequality $g(X_0^D(N)) \leq 34$ required for $X_0^D(N)$ to be tetragonal by Theorem 6.15. If this inequality is satisfied and $g(X_0^D(N)) \geq 2$, then we call such a pair a *tetragonal candidate*. All 516 tetragonal candidates are computed in `tetragonal_checks.m` and listed in the file `tetragonal_candidates.m` in [MPSS26]. We now narrow down these candidates.

Proposition 6.17.

(1) The curve $X_0^D(N)$ is geometrically tetragonal for the 133 pairs (D,N) listed in Table 9, for the 24 pairs listed in Table 10, and for the 4 pairs listed in Table 11.

(2) The curve $X_0^D(N)$ is tetragonal over $\mathbb{Q}$ for the 60 pairs in the set

$$\{(6,11),(6,17),(6,19),(6,23),(6,25),(6,29),(6,31),(6,37),(6,41),(6,71),(10,9),$$
$$(10,11),(10,13),(10,17),(10,23),(10,29),(14,3),(14,5),(14,13),(14,19),(15,2),$$
$$(15,4),(15,7),(15,13),(15,17),(21,2),(21,5),(21,11),(22,3),(22,5),(22,7),$$
$$(22,17),(26,1),(33,2),(33,7),(35,1),(35,2),(38,1),(39,1),(39,2),(39,4),(46,5),$$
$$(51,1),(51,2),(55,1),(58,1),(62,1),(69,1),(74,1),(86,1),(87,1),(94,1),(95,1),$$
$$(111,1),(119,1),(134,1),(146,1),(159,1),(194,1),(206,1)\}$$

and for the 81 pairs listed in Table 12.

(3) The curve $X_0^D(N)$ is not geometrically tetragonal (and, a fortiori, not tetragonal over $\mathbb{Q}$) for the 322 pairs (D,N) listed in Table 13, for the pair $(14,31)$, and for the 3 pairs in the set

$$\{(427,1),(445,1),(505,1)\}.$$

(4) The curve $X_0^D(N)$ is not tetragonal over $\mathbb{Q}$ for the 8 pairs (D,N) in the set

$$\{(51,7),(62,5),(65,2),(205,1),(217,1),(267,1),(301,1),(478,1)\}.$$

Proof. The code for all computations described in this proof can be found in the file `tetragonal_sieving.m` in [MPSS26].

(1) We witness the curves in Table 9 and Table 10 as being geometrically tetragonal via the existence of an Atkin–Lehner quotient map $X_0^D(N) \to X_0^D(N)/W$ with $W \leq W_0(D,N)$ of order 4 and with $g(X_0^D(N)/W) = 0$, or via the existence of such a map with $|W| = 2$ and with $g(X_0^D(N)/W) \leq 2$. All curves with at least one of these types of quotient maps appear in one of these tables, though we do not repeat pairs to be exhaustive in listing all such maps. For example, we only list $(6,25)$ in Table 9, but it also has an Atkin–Lehner subgroup of order 4 yielding a conic quotient curve.

We find that the four pairs listed in Table 11 are geometrically tetragonal via the existence of a sequence of covers

$$X_0^D(N) \to X_0^D(N)/\langle w_{m_1}\rangle \to X_0^D(N)/\langle w_{m_1}, w_{m_2}\rangle,$$

for $m_1 \neq m_2$ Hall divisors of DN, such that the bottom curve has genus 2 and the intermediate quotient has genus 3. It then follows from [Pol06, Theorem 3.4] that the intermediate quotient is geometrically hyperelliptic (and bielliptic), and so $X_0^D(N)$ is geometrically tetragonal. We list such values of m_1 and m_2 for each pair in Table 11.

(2) If (D,N) satisfies at least one of the following conditions, then the curve $X_0^D(N)$ is tetragonal over $\mathbb{Q}$:

(a) there is a subgroup $W \leq W_0(D,N)$ of order 4 so that $X_0^D(N)/W$ is isomorphic to $\mathbb{P}^1_{\mathbb{Q}}$, i.e., has genus 0 and a rational point;

(b) $X_0^D(N)$ is bielliptic over $\mathbb{Q}$;

(c) $X_0^D(N)$ has a degree 2 map to a curve which is sub-hyperelliptic over $\mathbb{Q}$.

If $X_0^D(N)$ is geometrically hyperelliptic (and hence, *a fortiori*, if it is hyperelliptic over $\mathbb{Q}$), then the hyperelliptic involution is an Atkin–Lehner involution w_m [Ogg83, p. 301]. Suppose that $X_0^D(N)$ is hyperelliptic over $\mathbb{Q}$ with hyperelliptic involution w_m, and take some involution $w_{m'} \neq w_m$. The quotient

$X_0^D(N)/\langle w_m, w_{m'}\rangle$, like $X_0^D(N)/\langle w_m\rangle$, must be isomorphic to $\mathbb{P}^1_{\mathbb{Q}}$; so in this case (D, N) satisfies the item (a) above and is tetragonal. By Part (3) of Proposition 6.14, this applies to the 33 pairs in the following set

$$\{(6,11),(6,19),(6,29),(6,31),(6,37),(10,11),(10,23),(14,5),(15,2),(22,3),(22,5),$$
$$(26,1),(35,1),(38,1),(39,1),(39,2),(51,1),(55,1),(58,1),(62,1),(69,1),(74,1),$$
$$(86,1),(87,1),(94,1),(95,1),(111,1),(119,1),(134,1),(146,1),(159,1),(194,1),(206,1)\}.$$

From [PS25a], the curve $X_0^D(N)$ is bielliptic over $\mathbb{Q}$, satisfying the item (b), for the following 23 pairs (D, N):

$$\{(6,17),(6,23),(6,25),(6,41),(6,71),(10,13),(10,17),(10,29),(14,3),(14,13),$$
$$(14,19),(15,7),(15,13),(15,17),(21,2),(21,5),(21,11),(22,7),(22,17),(33,2),$$
$$(33,7),(35,2),(46,5)\}.$$

The pairs $(15, 4)$ and $(39, 4)$ satisfy the item (c), as the corresponding curves are double covers of the curves $X_0^{15}(2)$ and $X_0^{39}(2)$, respectively, which are hyperelliptic over $\mathbb{Q}$. The pair $(51, 2)$ also satisfies the item (c); its quotient $X_0^{51}(2)/\langle w_{51}\rangle$ is bielliptic of genus 2 over $\mathbb{Q}$, with equation given in [PS25b]. The curve $X_0^{10}(9)$ is tetragonal by the item (a): its genus 0 quotient by the subgroup $\langle w_2, w_{45}\rangle$ has a rational point by [NR15, Proposition 4.25].

For the remaining pairs which we found to be geometrically tetragonal in Part (1) via Atkin–Lehner quotient maps, we attempt to show that $X_0^D(N)$ is tetragonal over $\mathbb{Q}$ by showing that either

- there is an Atkin–Lehner quotient map $X_0^D(N) \to X_0^D(N)/\langle w_m\rangle$ of degree 2 to a sub-hyperelliptic curve with a rational CM point, or
- there is an Atkin–Lehner quotient map $X_0^D(N) \to X_0^D(N)/\langle w_{m_1}, w_{m_2}\rangle$ of degree 4 with the latter quotient having genus 0 and a rational CM point, such that it is isomorphic to $\mathbb{P}^1_{\mathbb{Q}}$.

In both cases, we prove the existence of rational CM points on such quotients by using the work of [GR06] and [Sai24] to determine all quadratic CM points on $X_0^D(N)$, and using [GR06, Corollary 5.14] to determine such points whose image on a quotient of the form $X_0^D(N)/\langle w_m\rangle$ has residue field $\mathbb{Q}$. All 83 pairs handled in this manner, along with an Atkin–Lehner involution whose corresponding quotient provably has a rational CM point and is used in the argument, are listed in Table 12.

(3) For the 322 pairs listed in Table 13, we apply Lemma 6.8 (to the base change of $X_0^D(N)$ to $\overline{\mathbb{Q}}$) by finding a Hall divisor m of DN such that Equation (3) holds with $\sigma = w_m$ and such that $X_0^D(N)/\langle w_m\rangle$ is not geometrically sub-hyperelliptic. Such a quotient must have genus $g_m > 2$, as otherwise we would have found these curves $X_0^D(N)$ to be geometrically tetragonal in Part (1), and we prove such quotients are not geometrically hyperelliptic by one of the following means:
   (a) We have knowledge that this curve has no non-Atkin–Lehner automorphisms by the results of Section 5, and so it suffices to check for Atkin–Lehner quotients $X_0^D(N)/\langle w_m, w_{m'}\rangle$ with $m' > 1$ and $m' \neq m$ of genus 0.
   (b) We apply Lemma 6.6 to $X = X_0^D(N)/\langle w_m\rangle$ with $\sigma = w_{m'}$ for some Hall divisor $m' > 1$ of DN with $m' \neq m$.
   (c) We use Proposition 6.12, showing that $g_m \neq \frac{g}{2}$ or that $g_m \notin \left\{\frac{g-1}{2}, \frac{g+1}{2}\right\}$, for $g = g(X_0^D(N))$ and $g_m = g\left(X_0^D(N)/\langle w_m\rangle\right)$, according to whether g is even or odd.

(d) We use finite field point counts to show that the inequality in Lemma 6.11, Part (1), is not satisfied.

For the pair $(14, 31)$, we argue similarly using the Castelnuovo–Severi inequality with a larger Atkin–Lehner subgroup: $X_0^{14}(31)$ has a degree 4 map to the genus 1 curve $X_0^{14}(31)/\langle w_2, w_{217}\rangle$. If $X_0^{14}(31)$ were geometrically tetragonal, then a degree 4 map to a conic would be independent of the quotient map to $X_0^{14}(31)/\langle w_2, w_{217}\rangle$ unless both maps factored through a geometrically hyperelliptic curve $X_0^D(N)/\langle w_m\rangle$ for some $w_m \in \langle w_2, w_{217}\rangle$. We rule out this possibility similarly to the way we did in the previous paragraph. The Castelnuovo–Severi inequality then implies

$$17 = g(X_0^{14}(31)) \leq 4 \cdot 0 + 4 \cdot 1 + 3 \cdot 3 = 13,$$

giving a contradiction.

We now handle the last 3 mentioned pairs using Proposition 6.13. For each of these three curves $X = X_0^D(N)$, there exists a quotient $Y = Y_0^D(N)/\langle w_m\rangle$ of genus $g_Y \geq 10$ such that Y has a rational CM point. By Lemma 6.5 and Proposition 6.13, we then have

$$\gamma_{\mathbb{C}}(X) \geq \gamma_{\mathbb{C}}(Y) = \gamma_{\mathbb{Q}}(Y).$$

We then prove that X is not geometrically tetragonal by exhibiting such a curve Y for which point count computations and Lemma 3.3 prove that $\gamma_{\mathbb{Q}}(Y) > 4$. For example, the genus 15 quotient $Y = X_0^{505}(1)/\langle w_{505}\rangle$ of $X = X_0^{505}(1)$ has a rational -3-CM point and $Y(\mathbb{F}_4) = 24 > 20$, so $\gamma_{\mathbb{C}}(Y) > 4$.

(4) The 8 referenced curves $X_0^D(N)$ are proven to not be tetragonal over $\mathbb{Q}$ using point count computations and Lemma 3.3.

□

Proposition 6.18.

(1) For the 14 *triples* (D, N, W) *in the set*

$$\{(6, 55, \langle w_{10}, w_{66}\rangle), (6, 77, \langle w_{21}, w_{66}\rangle), (6, 83, \langle w_6, w_{166}\rangle), (10, 33, \langle w_{30}, w_{33}\rangle),$$
$$(14, 15, \langle w_{14}, w_{30}\rangle), (14, 23, \langle w_{14}, w_{46}\rangle), (21, 13, \langle w_{21}, w_{39}\rangle), (22, 19, \langle w_{22}, w_{38}\rangle),$$
$$(51, 5, \langle w_5, w_{51}\rangle), (55, 2, \langle w_2, w_{55}\rangle), (86, 3, \langle w_6, w_{86}\rangle), (87, 2, \langle w_2, w_{87}\rangle),$$
$$(95, 2, \langle w_{10}, w_{38}\rangle), (111, 2, \langle w_2, w_{111}\rangle)\},$$

we have that $X_0^D(N)$ *is tetragonal over* $\mathbb{Q}$ *if and only if* $\left(X_0^D(N)/W\right)(\mathbb{Q}) \neq \varnothing$.

(2) The curve $X_0^{46}(7)$ *is tetragonal over* $\mathbb{Q}$ *if and only if* $X_0^{46}(7)/\langle w_{161}\rangle$ *is hyperelliptic over* $\mathbb{Q}$.

Proof.

(1) For each (D, N, W) in the given set, we know $X_0^D(N)$ is geometrically tetragonal via the natural degree 4 map to the genus 0 curve $X_0^D(N)/W$. We also have $g(X_0^D(N)) \geq 10$, and we find that $X_0^D(N)$ is neither geometrically hyperelliptic nor geometrically bielliptic (either using finite field point count computations and Lemma 6.11 or using knowledge that all involutions are Atkin–Lehner from the results of Section 5). By Corollary 6.10, we then know that $X_0^D(N)$ is tetragonal over $\mathbb{Q}$ if and only if the known geometrically tetragonal map is tetragonal over $\mathbb{Q}$, i.e., if $X_0^D(N)/W \cong \mathbb{P}^1_{\mathbb{Q}}$.

(2) The genus 3 curve $Y = X_0^{46}(7)/\langle w_{161}\rangle$ is a degree 2 cover of the genus 2 curve $X_0^{46}(7)/\langle w_7, w_{23}\rangle$, and, hence, is geometrically hyperelliptic by [Pol06, Theorem 3.4]. We recognize that $X = X_0^{46}(7)$ is geometrically tetragonal via the

composition of the natural quotient map $X \to Y$ with the hyperelliptic quotient map $Y \to C$ to a conic over $\mathbb{Q}$, and note in particular that this composition is defined over $\mathbb{Q}$. As X has genus $13 \geq 10$ and is proven to be neither geometrically hyperelliptic nor geometrically bielliptic by either of the same checks referred to in the proof of Part (1), this composition is the unique geometrically tetragonal map on X by Lemma 6.9. Thus, X is tetragonal over $\mathbb{Q}$ if and only if the quotient of Y by its hyperelliptic involution has a rational point (as in Corollary 6.10). □

Corollary 6.19. *For the* 9 *pairs* (D, N) *in the set*

$$\{(6,55),(6,77),(6,83),(10,33),(14,15),(14,23),(21,13),(22,19),(86,3)\},$$

we have that $X_0^D(N)$ *is not tetragonal over* $\mathbb{Q}$.

Proof. For each pair (D, N) listed in the statement of the corollary, we know from Part (1) of Proposition 6.18 that $X_0^D(N)$ is tetragonal over $\mathbb{Q}$ if and only if the quotient $X_0^D(N)/W$ specified therein has a rational point. The claim then follows from [PS25b, Corollary 1.3]. (More specifically: using [PS25b, Algorithm 3.3], one can compute that there is at least one prime $p \mid D$ such that $X_0^D(N)/W$ has no $\mathbb{Q}_p$ points, giving the claim.) □

From the 516 tetragonal candidates, Proposition 6.17 narrows us down to only 29 pairs for which we remain unsure of whether $X_0^D(N)$ is geometrically tetragonal. These are listed in Table 1. Of course, if $X_0^D(N)$ is tetragonal over $\mathbb{Q}$, then it is also geometrically tetragonal, and so it must appear in Table 1 or in Table 9, Table 10, or Table 11. Of all such pairs, following Corollary 6.19, there remain 32 pairs for which we remain unsure of whether $X_0^D(N)$ is tetragonal over $\mathbb{Q}$. These pairs are listed in Table 2. Note that this latter table does not include all 32 of the pairs in Table 1; there are 8 curves which we prove are not tetragonal over $\mathbb{Q}$ using finite field point counts but for which we remain unsure of whether they are geometrically tetragonal, and likewise there are 11 curves X which are geometrically tetragonal but for which we remain unsure of whether X is tetragonal over $\mathbb{Q}$. This completes the proof of Theorem 1.5.

TABLE 1. All 29 pairs (D, N) for which we remain unsure of whether $X_0^D(N)$ is geometrically tetragonal following Theorem 1.5. Pairs for which we have proved that $X_0^D(N)$ is not tetragonal over $\mathbb{Q}$ are written in bold.

(D,N)	g	(D,N)	g	(D,N)	g	(D,N)	g
$(6,73)$	11	$(10,27)$	13	$(15,16)$	17	$(22,9)$	11
$(33,4)$	11	$(34,13)$	17	$(38,5)$	9	$(46,9)$	23
$\mathbf{(51,7)}$	**21**	$(55,3)$	13	$\mathbf{(62,5)}$	**15**	$\mathbf{(65,2)}$	**13**
$(69,2)$	11	$(94,3)$	15	$(133,1)$	9	$(177,1)$	9
$(187,1)$	13	$\mathbf{(205,1)}$	**13**	$(213,1)$	11	$(214,1)$	8
$\mathbf{(217,1)}$	**15**	$(226,1)$	9	$(262,1)$	10	$\mathbf{(267,1)}$	**15**
$(298,1)$	12	$\mathbf{(301,1)}$	**21**	$(358,1)$	14	$(382,1)$	15
$\mathbf{(478,1)}$	**19**						

TABLE 2. All 32 pairs (D, N) for which we remain unsure of whether $X_0^D(N)$ is tetragonal over $\mathbb{Q}$ following Theorem 1.5. Pairs for which we have proved that $X_0^D(N)$ is geometrically tetragonal are written in bold.

(D,N)	g	(D,N)	g	(D,N)	g	(D,N)	g

(6, 49)	**9**	(6, 73)	11	(10, 27)	13	**(14, 9)**	**7**
(15, 8)	**9**	(15, 16)	17	**(21, 4)**	**7**	**(21, 8)**	**13**
(22, 9)	11	(33, 4)	11	(34, 13)	17	(38, 5)	9
(46, 7)	**13**	(46, 9)	23	**(51, 5)**	**17**	**(55, 2)**	**11**
(55, 3)	13	(69, 2)	11	**(87, 2)**	**15**	(94, 3)	15
(95, 2)	**19**	**(111, 2)**	**19**	(133, 1)	9	(177, 1)	9
(187, 1)	13	(213, 1)	11	(214, 1)	8	(226, 1)	9
(262, 1)	10	(298, 1)	12	(358, 1)	14	(382, 1)	15

The pairs in Table 1 and Table 2 appear to require further techniques to handle, and we leave them to future work. In particular, those pairs in Table 1 with N not squarefree are not amenable to our techniques for determining the automorphism group from Section 5, and those pairs with genus $g \leq 9$ are not amenable to Corollary 6.10.

7. Point count records and maximal curves

In this section, we display tables with our best results on the number of points for a curve of given genus over a specified finite field among Atkin–Lehner quotients of Shimura curves $X_0^D(N)$.

Table 3 collects 116 improvements with respect to data collected on [vdGHLR09]. These improvements have been shared with the authors of the referenced site and are included in their database at the time of writing this work. They are all (non-trivial) Atkin–Lehner quotients of Shimura curves $X_0^D(N)$.

Table 3. The 116 improved bounds for $|X(\mathbb{F}_{p^k})|$ among curves X of the form $X_0^D(N)/\langle w_{m_1}, \ldots, w_{m_r}\rangle$ for $DN \leq 10000$, $D > 1$, genus $g \leq 50$, primes $p \leq 19$, and $k \leq 5$.

g	p^k	(D, N)	$m_1, \ldots, m_r$	$\lvert X(\mathbb{F}_{p^k})\rvert$	g	p^k	(D, N)	$m_1, \ldots, m_r$	$\lvert X(\mathbb{F}_{p^k})\rvert$
8	3^2	(14, 95)	5, 266	44	8	13^2	(6, 209)	57, 66	370
8	17^5	(546, 5)	2, 91, 39	1436014	10	7^2	(10, 291)	6, 5, 97	170
10	7^5	(38, 15)	3, 19	18850	10	17^5	(14, 141)	2, 7, 47	1441950
10	19^2	(215, 4)	5, 43	720	11	2^2	(4485, 1)	3, 13, 115	28
11	5^5	(21, 67)	7, 67	4000	11	7^2	(6474, 1)	3, 83, 26	184
11	7^5	(4182, 1)	17, 6, 123	18883	11	19^2	(39, 34)	6, 13, 17	748
12	7^2	(26, 57)	57, 78	190	12	7^5	(6, 475)	25, 57, 6	19682
12	11^2	(15, 178)	2, 89, 15	342	12	13^5	(6, 265)	265, 30	381782
12	17^5	(570, 11)	5, 6, 209	1441318	13	7^2	(46, 25)	2, 23	186
13	7^5	(82, 15)	41, 10	19456	13	11^5	(39, 8)	3	168744
13	13^2	(10, 99)	10, 22	456	13	13^4	(35, 9)	7	32436
13	17^5	(2415, 1)	15, 161	1444848	14	3^2	(65, 14)	5, 91	60
14	7^2	(10, 97)	97	198	14	13^2	(714, 11)	33, 238, 1309	460
14	13^5	(6, 265)	30, 159	384006	14	19^5	(134, 15)	2, 5, 67	2514998
15	13^5	(15, 112)	16, 5, 21	386416	15	17^2	(1590, 1)	30	724
17	7^5	(82, 15)	2, 41	20106	17	13^2	(10, 63)	10	544
17	13^5	(15, 32)	5	386880	18	7^2	(22, 43)	43	230
18	17^2	(26, 57)	3, 247	784	19	7^5	(26, 27)	13	20520

19	17^5	$(14, 141)$	$14, 94$	1451936	20	19^2	$(14, 117)$	$13, 63$	946
21	5^2	$(14, 277)$	$7, 554$	164	21	11^2	$(57, 14)$	399	476
21	13^2	$(10, 171)$	$10, 19$	596	21	17^2	$(2415, 2)$	$15, 14, 46$	848
22	2^2	$(15, 197)$	$3, 985$	45	22	3^2	$(215, 4)$	215	84
22	11^2	$(14, 135)$	$7, 270$	498	22	13^2	$(57, 28)$	$19, 21$	594
22	19^2	$(51, 26)$	$39, 51$	1040	23	2^2	$(39, 31)$	1209	46
23	11^2	$(35, 26)$	455	508	23	13^2	$(22, 49)$	22	640
23	17^2	$(22, 87)$	$3, 58$	938	23	19^2	$(690, 7)$	$14, 483$	1004
24	5^2	$(14, 253)$	$7, 506$	190	24	17^2	$(2415, 2)$	$161, 69, 115$	910
25	13^2	$(10, 99)$	22	696	25	19^2	$(34, 123)$	$3, 17, 41$	1091
26	11^2	$(21, 118)$	$14, 1239$	550	26	17^2	$(6, 539)$	$2, 147$	928
26	19^2	$(26, 85)$	$13, 34$	1106	27	13^2	$(4935, 1)$	$5, 987$	686
27	17^2	$(3045, 2)$	$58, 435, 1015$	978	28	5^2	$(38, 27)$	19	216
28	11^2	$(35, 52)$	$5, 91$	570	28	13^2	$(15, 164)$	$15, 41$	756
28	19^2	$(33, 58)$	$11, 29$	1152	29	3^2	$(35, 89)$	$5, 623$	108
29	5^2	$(623, 3)$	$3, 623$	208	29	11^2	$(65, 14)$	455	572
29	13^2	$(33, 56)$	$11, 21$	704	29	17^2	$(15, 77)$	33	1020
29	19^2	$(10, 117)$	130	1216	30	2^2	$(1169, 1)$	1169	55
30	3^2	$(391, 4)$	$17, 23$	109	30	13^2	$(118, 9)$	118	746
30	17^2	$(177, 8)$	$3, 59$	1050	31	11^2	$(14, 345)$	$5, 23, 14$	684
31	13^2	$(15, 62)$	93	780	31	19^2	$(1155, 2)$	$7, 11$	1272
32	3^2	$(1055, 2)$	$2, 1055$	113	32	5^2	$(77, 37)$	$7, 407$	221
32	11^2	$(15, 178)$	$15, 89$	628	32	19^2	$(33, 62)$	$11, 93$	1278
33	5^2	$(57, 31)$	1767	236	33	13^2	$(1055, 2)$	$422, 10$	773
33	17^2	$(1055, 2)$	$10, 422$	1103	34	3^2	$(871, 2)$	$2, 871$	121
34	13^2	$(10, 297)$	$22, 297$	810	34	19^2	$(134, 9)$	67	1470
36	5^2	$(1343, 1)$	1343	250	36	11^2	$(142, 9)$	142	698
37	5^2	$(327, 4)$	327	264	37	13^2	$(145, 23)$	$5, 667$	846
38	2^2	$(15, 611)$	$3, 13, 235$	65	38	5^2	$(3143, 1)$	$7, 449$	257
39	2^2	$(65, 31)$	2015	68	39	5^2	$(3927, 2)$	$2, 231, 17$	266
40	5^2	$(287, 4)$	287	282	40	11^2	$(91, 20)$	$13, 35$	762
40	19^2	$(10, 351)$	$270, 351$	1488	41	5^2	$(143, 6)$	143	280
42	3^2	$(58, 97)$	$2, 2813$	134	43	2^2	$(15, 301)$	$5, 903$	74
43	5^2	$(69, 22)$	759	284	43	19^2	$(21, 230)$	$15, 161, 230$	1524
44	2^2	$(1589, 1)$	1589	71	44	3^2	$(185, 19)$	$5, 703$	140
45	5^2	$(623, 2)$	623	308	45	7^2	$(1751, 1)$	1751	456
46	7^2	$(695, 2)$	695	472	47	2^2	$(7215, 1)$	$5, 1443$	78
47	7^2	$(15, 146)$	1095	472	47	19^2	$(21, 62)$	93	1624
48	5^2	$(1623, 2)$	$2, 1623$	307	48	19^2	$(51, 52)$	$39, 51$	1700
49	17^2	$(35, 24)$	21	1488	50	5^2	$(57, 62)$	$2, 1767$	330

In Tables 4, 5, and 6, we list Atkin–Lehner quotients $X_0^D(N)/\langle w_{d_1}, \ldots, w_{d_k}\rangle$ with genus $g \geq 2$ achieving the maximal number of points for a certain pair (genus, finite

field). The curves are grouped by their real Weil polynomial $h_W(x)$ which is displayed as well.

We write in bold curves that belong to classes sharing a common real Weil polynomial that does not occur among those previously encountered in this paper or in [DLMS26, Appendix C, Tables 11, 12, and 13]. This implies the existence of a previously unknown maximal isomorphism classes.

We next explain the notation of the tables which we use to save space. Let

$$DN = \prod_i p_i^{v_i} = \prod_i q_i$$

be the prime factorization of DN, with $p_1 < p_2 < \dots$ and $q_i = p_i^{v_i}$. Then, each Hall divisor d_h is uniquely the product of certain q_j's and, in the tables, we denote $X_0^D(N)/\langle w_{d_1}, \dots, w_{d_k}\rangle$ by

$$(D, N)\,\{i_{1,1}, \dots, i_{1,m_1}; \dots\dots\dots; i_{k,1}, \dots, i_{k,m_k}\},$$

where

$$d_h = \prod_{j=1}^{m_h} q_{i_{h,j}}, \qquad \text{for } h = 1, \dots, k.$$

The top curve $X_0^D(N)$ is denoted by $(D, N)\,\{\}$ in this notation. For a non-trivial example, the entry "$(10, 27)\{2; 1, 3\}$" in the table below, corresponding to the third curve in the second class of the row $g = 2$ and $q = 7$, denotes the quotient of the curve $X_0^{10}(27)$ by $\langle w_{27}, w_{10}\rangle$, where $d_1 = 27 = p_2^3$ and $d_2 = 10 = p_1 p_3$ since, in this case, $p_1 = 2$, $p_2 = 3$, and $p_3 = 5$.

Another example is the entry "$(462, 5)\{2; 3; 1, 5; 4, 5\}$" in the table below, corresponding to the first curve in the row $g = 2$ and $q = 17^3$, that denotes the quotient of the curve $X_0^{462}(5)$ by $\langle w_3, w_5, w_{22}, w_{77}\rangle$, where $d_1 = 3 = p_2$ and $d_2 = 5 = p_3$ and $d_3 = 22 = p_1 p_5$ and $d_4 = 77 = p_4 p_5$ since, in this case, $p_1 = 2$, $p_2 = 3$, $p_3 = 5$, $p_4 = 7$, and $p_5 = 11$.

TABLE 4. 584 maximal curves over $\mathbb{F}_q$ of genus $g = 2$.

g	q	$h_W(x)$	Curves
2	2	$x^2 + 3x + 1$	$(15, 37)\{1; 2; 3\}$, $(21, 23)\{1; 2; 3\}$, $(93, 1)\{1, 2\}$, $(115, 1)\{1, 2\}$, $(177, 1)\{1; 2\}$, $(213, 1)\{1; 2\}$, $(1365, 1)\{1; 2; 3; 4\}$
2	2^2	$x^2 + 5x + 9$	$(15, 29)\{1; 2, 3\}$, $(15, 73)\{1; 2; 3\}$, $(15, 91)\{1; 2; 3; 4\}$, $(21, 11)\{1; 2\}$, $(21, 43)\{1; 2; 3\}$, $(33, 7)\{1; 3\}$, $(35, 39)\{1; 2; 3; 4\}$, $(77, 3)\{2; 3\}$, $(87, 1)\{1\}$, $(87, 5)\{1; 2; 3\}$, $(111, 5)\{1; 2; 3\}$, $(161, 1)\{1, 2\}$, $(161, 3)\{1; 2; 3\}$, $(265, 1)\{1; 2\}$, $(287, 1)\{1; 2\}$, $(291, 1)\{1; 2\}$, $(299, 1)\{1; 2\}$, $(321, 1)\{1; 2\}$, $(341, 1)\{1; 2\}$, $(371, 1)\{1; 2\}$, $(415, 1)\{1; 2\}$, $(417, 1)\{1; 2\}$, $(453, 1)\{1; 2\}$, $(471, 1)\{1; 2\}$
2	2^7	$x^2 + 43x + 713$	$(15, 29)\{1; 2, 3\}$, $(15, 73)\{1; 2; 3\}$, $(15, 91)\{1; 2; 3; 4\}$, $(21, 43)\{1; 2; 3\}$, $(35, 39)\{1; 2; 3; 4\}$, $(87, 5)\{1; 2; 3\}$, $(111, 5)\{1; 2; 3\}$, $(161, 1)\{1, 2\}$, $(265, 1)\{1; 2\}$, $(287, 1)\{1; 2\}$, $(291, 1)\{1; 2\}$, $(299, 1)\{1; 2\}$, $(321, 1)\{1; 2\}$, $(371, 1)\{1; 2\}$, $(415, 1)\{1; 2\}$, $(417, 1)\{1; 2\}$, $(453, 1)\{1; 2\}$, $(471, 1)\{1; 2\}$
2	3	$x^2 + 4x + 2$	$(10, 67)\{1; 2; 3\}$, $(85, 1)\{1, 2\}$, $(118, 5)\{1; 2; 3\}$
		$x^2 + 4x + 3$	$(10, 19)\{1, 2, 3\}$, $(10, 47)\{1; 2, 3\}$, $(10, 47)\{2; 1, 3\}$, $(10, 83)\{1; 2; 3\}$, $(26, 11)\{1; 2, 3\}$, $(26, 19)\{1; 2, 3\}$, $(38, 5)\{1; 2\}$, $(38, 5)\{1; 2, 3\}$, $(38, 5)\{2; 1, 3\}$, $(58, 1)\{\}$, $(142, 1)\{1, 2\}$, $(142, 5)\{1; 2; 3\}$, $(158, 1)\{1, 2\}$, $(302, 1)\{1; 2\}$, $(1430, 1)\{1; 4; 2, 3\}$, $(1430, 1)\{3; 4; 1, 2\}$, $(1870, 1)\{1; 2; 3; 4\}$
		$x^2 + 4x + 4$	$(10, 43)\{1; 2, 3\}$, $(10, 91)\{2; 3; 1, 4\}$, $(14, 5)\{3\}$, $(14, 17)\{2; 1, 3\}$, $(14, 31)\{2; 1, 3\}$, $(14, 55)\{2; 3; 1, 4\}$, $(22, 23)\{3; 1, 2\}$, $(34, 5)\{1, 2, 3\}$, $(34, 7)\{1; 2, 3\}$, $(65, 2)\{1; 2, 3\}$, $(145, 1)\{1; 2\}$, $(145, 2)\{1; 2; 3\}$, $(770, 1)\{1, 3; 2, 4\}$, $(1330, 1)\{1; 2; 3, 4\}$
2	3^2	$x^2 + 10x + 37$	$(10, 11)\{1\}$, $(10, 11)\{1, 2\}$, $(10, 11)\{1, 3\}$, $(10, 17)\{3\}$, $(10, 19)\{1, 2\}$, $(10, 71)\{1; 2, 3\}$, $(10, 71)\{2; 1, 3\}$, $(10, 73)\{1; 2, 3\}$, $(10, 77)\{1, 2; 3, 4\}$, $(10, 119)\{1; 4; 2, 3\}$, $(10, 119)\{2; 4; 1, 3\}$, $(10, 121)\{1; 2; 3\}$, $(10, 137)\{1; 2; 3\}$, $(14, 13)\{1, 2\}$, $(14, 61)\{1; 2, 3\}$, $(14, 61)\{3; 1, 2\}$, $(14, 65)\{1; 3; 2, 4\}$, $(14, 65)\{1, 3; 2, 4; 3, 4\}$, $(14, 109)\{1; 2; 3\}$, $(22, 5)\{1\}$, $(22, 5)\{2\}$, $(22, 5)\{3\}$, $(22, 13)\{1, 2, 3\}$, $(22, 25)\{1; 2; 3\}$, $(26, 7)\{2; 3\}$, $(26, 11)\{2; 1, 3\}$,

			$(26,11)\{3;1,2\}$, $(26,19)\{2;1,3\}$, $(26,19)\{3;1,2\}$, $(26,29)\{1;2;3\}$, $(26,55)\{1;2;3;4\}$, $(34,23)\{1;2;3\}$, $(35,2)\{3\}$, $(35,4)\{1;3\}$, $(35,19)\{1;2;3\}$, $(38,1)\{\}$, $(38,7)\{1,2,3\}$, $(38,11)\{2;3\}$, $(38,11)\{1;2,3\}$, $(38,13)\{1;2,3\}$, $(46,19)\{1;2;3\}$, $(55,2)\{2,3\}$, $(55,4)\{1;2,3\}$, $(55,4)\{1,3;2,3\}$, $(58,7)\{1;2,3\}$, $(58,7)\{3;1,2\}$, $(62,13)\{1;2;3\}$, $(82,7)\{3;1,2\}$, $(94,5)\{2;1,3\}$, $(94,7)\{1,3;2,3\}$, $(95,2)\{2,3\}$, $(95,4)\{1;2,3\}$, $(106,5)\{2;1,3\}$, $(115,1)\{1,2\}$, $(115,2)\{1;2,3\}$, $(143,2)\{1;2,3\}$, $(155,1)\{1,2\}$, $(161,1)\{1,2\}$, $(203,2)\{1;2;3\}$, $(235,2)\{1;2;3\}$, $(341,1)\{1;2\}$, $(394,1)\{1;2\}$, $(446,1)\{1,2\}$, $(482,1)\{1;2\}$, $(502,1)\{1;2\}$, $(586,1)\{1;2\}$, $(626,1)\{1;2\}$, $(1190,1)\{2;1,3,4\}$, $(2090,1)\{2;3;1,4\}$, $(3190,1)\{1;2;3;4\}$, $(3542,1)\{1;2;3;4\}$
2	3^3	$x^2+20x+148$	$(14,29)\{1;3\}$, $(77,2)\{1;3\}$
2	3^4	$x^2+36x+480$	$(10,31)\{2;3\}$, $(14,25)\{1,3;2,3\}$
2	3^5	$x^2+62x+1441$	$(10,71)\{1;2,3\}$, $(10,71)\{2;1,3\}$, $(10,73)\{1;2,3\}$, $(10,121)\{1;2;3\}$, $(10,137)\{1;2;3\}$, $(14,61)\{1;2,3\}$, $(14,109)\{1;2;3\}$, $(22,13)\{1,2,3\}$, $(22,25)\{1;2;3\}$, $(26,11)\{2;1,3\}$, $(26,19)\{3;1,2\}$, $(26,29)\{1;2;3\}$, $(26,55)\{1;2;3;4\}$, $(34,23)\{1;2;3\}$, $(35,19)\{1;2;3\}$, $(38,11)\{1;2,3\}$, $(38,13)\{1;2,3\}$, $(58,7)\{1;2,3\}$, $(62,13)\{1;2;3\}$, $(106,5)\{2;1,3\}$, $(115,1)\{1,2\}$, $(115,2)\{1;2,3\}$, $(143,2)\{1;2,3\}$, $(155,1)\{1,2\}$, $(161,1)\{1,2\}$, $(235,2)\{1;2;3\}$, $(341,1)\{1;2\}$, $(394,1)\{1;2\}$, $(446,1)\{1,2\}$, $(482,1)\{1;2\}$, $(502,1)\{1;2\}$, $(586,1)\{1;2\}$, $(626,1)\{1;2\}$, $(2090,1)\{2;3;1,4\}$, $(3190,1)\{1;2;3;4\}$, $(3542,1)\{1;2;3;4\}$
2	5	x^2+6x+7	$(\mathbf{249},\mathbf{1})\{\mathbf{1};\mathbf{2}\}$, $(\mathbf{2562},\mathbf{1})\{\mathbf{1};\mathbf{2};\mathbf{3};\mathbf{4}\}$
		x^2+6x+8	$(6,17)\{1,2\}$, $(6,49)\{1,2,3\}$, $(6,103)\{1,2,3\}$, $(6,133)\{2;4;1,3\}$, $(6,133)\{2;3;1,4\}$, $(6,157)\{1;2;3\}$, $(6,173)\{1;2;3\}$, $(14,17)\{3;1,2\}$, $(14,33)\{1;2;3,4\}$, $(22,9)\{3;1,2\}$, $(33,4)\{3;1,2\}$, $(38,11)\{1;2,3\}$, $(51,2)\{2,3\}$, $(51,4)\{1;2,3\}$, $(123,1)\{1,2\}$, $(142,1)\{1,2\}$, $(462,1)\{3;1,2,4\}$, $(798,1)\{1;2,3,4\}$
		x^2+6x+9	$(6,61)\{1,2,3\}$, $(6,199)\{1;2;3\}$, $(26,11)\{3;1,2\}$, $(26,33)\{1;2;3;4\}$, $(58,7)\{3;1,2\}$, $(91,1)\{1,2\}$, $(91,2)\{1;2,3\}$, $(133,1)\{1;2\}$, $(133,2)\{1;2;3\}$, $(177,1)\{1;2\}$, $(177,2)\{1;2;3\}$, $(321,1)\{1;2\}$, $(1482,1)\{1;2;3,4\}$, $(1554,1)\{1;2;3,4\}$, $(1806,1)\{1;2;3,4\}$, $(2814,1)\{1;2;3;4\}$
2	5^2	$x^2+20x+140$	$(6,19)\{1,2\}$, $(6,37)\{1\}$, $(6,73)\{1;2\}$, $(6,77)\{4;1,2\}$, $(6,77)\{2;3;4\}$, $(6,143)\{2;3;1,4\}$, $(14,3)\{3\}$, $(14,9)\{2;3\}$, $(14,13)\{2;3\}$, $(14,23)\{2;3\}$, $(14,37)\{1;3\}$, $(14,39)\{3;4;1,2\}$, $(21,8)\{2;3\}$, $(21,13)\{2;3\}$, $(21,26)\{3;4;1,2\}$, $(26,7)\{1,3\}$, $(26,21)\{2,3;1,4\}$, $(34,3)\{1,2\}$, $(34,3)\{2,3\}$, $(34,7)\{1;2\}$, $(34,11)\{1,2,3\}$, $(38,3)\{2;3\}$, $(39,7)\{1,3;2,3\}$, $(39,8)\{1;2,3\}$, $(51,26)\{1;2;3;4\}$, $(57,2)\{3\}$, $(57,4)\{1;3\}$, $(69,2)\{1;2\}$, $(111,2)\{2,3\}$, $(111,4)\{1;2,3\}$, $(122,1)\{2\}$, $(237,1)\{2\}$, $(462,1)\{2;1,3\}$, $(462,1)\{1,3;1,4\}$, $(1122,1)\{3;4;1,2\}$, $(1326,1)\{1,4;2,3\}$, $(1938,1)\{3;1,2;2,4\}$, $(1974,1)\{2;3;1,4\}$
2	5^3	$x^2+44x+724$	$(6,97)\{2;3\}$, $(6,113)\{1,3;2,3\}$, $(6,143)\{1;3;4\}$, $(6,203)\{1;2,3;3,4\}$, $(22,19)\{2;3\}$, $(22,21)\{1;2;3\}$, $(22,21)\{2;3;4\}$, $(39,2)\{1;3\}$, $(1122,1)\{1;2;4\}$, $(1122,1)\{1;3;4\}$, $(1218,1)\{1;3;4\}$
2	7	$x^2+8x+15$	$(6,43)\{1,2,3\}$, $(6,101)\{2;1,3\}$, $(6,127)\{1;2,3\}$, $(10,43)\{2;1,3\}$, $(15,23)\{1;2,3\}$, $(122,3)\{2;1,3\}$, $(1518,1)\{1;3;2,4\}$
		$x^2+8x+16$	$(15,4)\{2\}$, $(10,13)\{1,3\}$, $(10,27)\{2;1,3\}$, $(10,33)\{2,3;1,2,4\}$, $(22,15)\{2;1,3,4\}$, $(34,15)\{1;2;3,4\}$, $(65,2)\{1;2,3\}$, $(1830,1)\{1;2;3,4\}$
2	7^2	$x^2+28x+280$	$(6,55)\{1;2,4\}$, $(6,59)\{1;3\}$, $(6,83)\{1;3\}$, $(6,85)\{1;2;3\}$, $(6,95)\{1;3;4\}$, $(6,115)\{1;2;4\}$, $(6,115)\{1;3;4\}$, $(6,143)\{1;2,3;3,4\}$, $(10,39)\{1;2;3\}$, $(10,69)\{1;3;4\}$, $(15,2)\{3\}$, $(15,4)\{1;3\}$, $(15,8)\{1;3\}$, $(15,8)\{3;1,2\}$, $(15,22)\{2,3;2,4\}$, $(15,22)\{3,4;1,2,3\}$, $(15,22)\{3;1,2;2,4\}$, $(15,26)\{3;1,2;2,4\}$, $(15,32)\{1;3;2\}$, $(15,34)\{3;1,2;2,4\}$, $(15,44)\{3;4;1,2\}$, $(15,44)\{1;2,3;3,4\}$, $(22,43)\{1;2;3\}$, $(26,15)\{1,4;2,3\}$, $(26,15)\{1;2;4\}$, $(51,1)\{2\}$, $(51,2)\{1;3\}$, $(51,2)\{3;1,2\}$, $(51,5)\{3;1,2\}$, $(51,10)\{2;3;4\}$, $(51,10)\{1;4;2,3\}$, $(51,20)\{1;2;3;4\}$, $(330,1)\{1;3\}$, $(330,1)\{1,3;2,3\}$, $(330,1)\{4;1,3\}$, $(330,13)\{1;3;2,4;5\}$, $(690,1)\{1;2,4\}$, $(690,1)\{1;2;3\}$, $(858,1)\{1;3;4\}$, $(1290,1)\{3;1,2;2,4\}$
2	11	$x^2+12x+36$	$(21,8)\{2;3\}$
2	11^2	$x^2+44x+704$	$(6,17)\{3\}$, $(6,25)\{3\}$, $(6,35)\{2;1,3\}$, $(6,35)\{2;3,4\}$, $(6,65)\{2;1,3\}$, $(6,65)\{2;3;4\}$, $(6,65)\{3;1,2,4\}$, $(6,67)\{1,2,3\}$, $(6,79)\{1,3;2,3\}$, $(6,85)\{1;4\}$, $(6,85)\{2;3;4\}$, $(6,85)\{4;1,2,3\}$, $(6,95)\{2;3;4\}$, $(6,95)\{1;2,3;3,4\}$, $(6,115)\{1;3;4\}$, $(6,115)\{2;3;4\}$, $(6,145)\{3,4;1,2,3\}$, $(6,203)\{3;4;1,2\}$, $(10,17)\{3\}$, $(10,19)\{1,2\}$, $(10,21)\{2,3;2,4\}$, $(10,21)\{1,2;1,3,4\}$, $(10,21)\{2,3;1,2,4\}$, $(10,27)\{2;1,3\}$, $(10,39)\{4;1,3;2,3\}$, $(10,41)\{1,3;2,3\}$, $(10,51)\{4;1,3\}$, $(10,51)\{4;1,2,3\}$, $(10,51)\{1;2;4\}$, $(10,53)\{2;1,3\}$, $(14,3)\{3\}$, $(14,5)\{3\}$, $(14,9)\{2;3\}$, $(14,15)\{4;1,2,3\}$, $(14,15)\{4;1,3;2,3\}$, $(14,23)\{3;1,2\}$, $(14,115)\{1;2;3;4\}$, $(15,4)\{2\}$, $(15,14)\{2;1,3,4\}$, $(15,14)\{1;2;3\}$, $(15,34)\{4;2,3\}$, $(15,68)\{1;4;2,3\}$, $(21,10)\{1;2;4\}$, $(26,15)\{1,4;2,3\}$, $(26,45)\{1;2;3;4\}$, $(35,6)\{1;3;4\}$, $(35,18)\{1;2;3;4\}$, $(62,5)\{1;2,3\}$, $(62,7)\{1,3;2,3\}$, $(86,1)\{2\}$, $(95,2)\{2,3\}$, $(95,4)\{1;2,3\}$, $(142,1)\{1\}$, $(142,3)\{1;2,3\}$, $(210,1)\{2;4\}$, $(210,13)\{2;3;1,5;4,5\}$, $(210,17)\{1;4;5;2,3\}$, $(210,17)\{2,4;3,5;1,2,5\}$, $(390,1)\{2;4\}$, $(390,1)\{2;1,4\}$, $(390,1)\{2;3,4\}$, $(390,7)\{2;5;1,3,4\}$, $(390,7)\{1;2;3,5;4,5\}$, $(390,7)\{2;1,5;3,5;4,5\}$, $(510,1)\{3;1,2;2,4\}$, $(570,1)\{2;3,4\}$, $(570,1)\{1;3;4\}$, $(966,1)\{2;1,3;3,4\}$, $(1218,1)\{1,3;2,4;3,4\}$, $(1290,1)\{2;1,3;3,4\}$, $(1794,1)\{1;4;2,3\}$, $(2130,1)\{1;3;2,4\}$
2	11^3	$x^2+144x+7824$	$(6,91)\{2;3,4\}$, $(6,155)\{3;1,2;2,4\}$, $(10,37)\{2,3\}$, $(546,1)\{2;3,4\}$, $(930,1)\{1,2;1,3,4\}$

2	11^5	$x^2+1604x+965284$	$(6,67)\{1,3;2,3\}$, $(6,73)\{1,3;2,3\}$, $(6,101)\{2;1,3\}$, $(6,197)\{1;2;3\}$, $(6,215)\{2;4;1,3\}$, $(6,221)\{1;3;2,4\}$, $(6,221)\{2;3;1,4\}$, $(6,229)\{1;2;3\}$, $(14,31)\{3;1,2\}$, $(14,41)\{1;2;3\}$, $(26,3)\{1\}$, $(26,5)\{1;2\}$, $(26,15)\{1;3;2,4\}$, $(26,17)\{1;2,3\}$, $(26,23)\{1;2;3\}$, $(26,43)\{1;2;3\}$, $(34,39)\{1;2;3;4\}$, $(39,1)\{1\}$, $(39,2)\{1;2\}$, $(39,2)\{2;1,3\}$, $(39,4)\{2;1,3\}$, $(39,5)\{1;2,3\}$, $(39,7)\{1;2,3\}$, $(39,10)\{1;2;3,4\}$, $(46,35)\{1;2;3;4\}$, $(51,26)\{1;2;3;4\}$, $(194,5)\{1;2;3\}$, $(210,13)\{3,5;1,2,5;2,4,5\}$, $(210,17)\{2;1,5;3,5;4,5\}$, $(321,1)\{1;2\}$, $(415,1)\{1;2\}$, $(510,7)\{2;4;5;1,3\}$, $(798,1)\{3;1,2;2,4\}$, $(1190,1)\{2;1,3,4\}$, $(1302,1)\{4;1,3;2,3\}$, $(1326,1)\{2;1,3,4\}$, $(1610,1)\{1;4;2,3\}$, $(1770,1)\{1,4;2,3\}$, $(2886,1)\{1;2;3;4\}$, $(4326,1)\{1;2;3;4\}$
2	13	$x^2+12x+35$	$(158,1)\{1,2\}$, $(1518,1)\{1;2;3,4\}$, $(1830,1)\{2;4;1,3\}$
		$x^2+12x+36$	$(6,77)\{1;4;2,3\}$, $(14,25)\{1;2,3\}$, $(22,21)\{1;2;3,4\}$, $(35,6)\{3,4;1,2,3\}$, $(155,1)\{1,2\}$, $(210,11)\{1,5;2,5;3,4,5\}$, $(690,1)\{4;1,2,3\}$, $(714,5)\{1;3;5;2,4\}$
2	13^2	$x^2+52x+988$	$(6,29)\{1,2\}$, $(6,79)\{1,3;2,3\}$, $(6,95)\{1,3;2,4;3,4\}$, $(58,3)\{3\}$, $(82,1)\{1\}$, $(82,3)\{1;2,3\}$, $(210,17)\{2;3;1,5;4,5\}$, $(249,1)\{1;2\}$, $(566,1)\{1;2\}$, $(570,1)\{1,2;3,4\}$, $(690,1)\{4;1,2\}$, $(870,1)\{1;2;4\}$, $(870,1)\{4;1,3;2,3\}$, $(1122,1)\{2;3;4\}$, $(1122,1)\{4;1,3;2,3\}$, $(1230,1)\{1;2,3;3,4\}$, $(1254,1)\{4;1,3;2,3\}$, $(1914,1)\{2;4;1,3\}$
2	17^2	$x^2+68x+1700$	$(\mathbf{6,43})\{\mathbf{1,2,3}\}$, $(\mathbf{6,49})\{\mathbf{1,3;2,3}\}$, $(\mathbf{6,65})\{\mathbf{2;1,3}\}$, $(\mathbf{6,115})\{\mathbf{3;4;1,2}\}$, $(\mathbf{6,143})\{\mathbf{2;1,3;3,4}\}$, $(\mathbf{6,155})\{\mathbf{2;1,3;3,4}\}$, $(\mathbf{6,161})\{\mathbf{3;4;1,2}\}$, $(\mathbf{6,161})\{\mathbf{4;1,3;2,3}\}$, $(\mathbf{6,235})\{\mathbf{3;1,2;2,4}\}$, $(\mathbf{10,93})\{\mathbf{3;1,2;2,4}\}$, $(\mathbf{14,19})\{\mathbf{1;2}\}$, $(\mathbf{14,57})\{\mathbf{1;3;2,4}\}$, $(\mathbf{15,11})\{\mathbf{1;3}\}$, $(\mathbf{15,19})\{\mathbf{1;2}\}$, $(\mathbf{15,22})\{\mathbf{2;4;1,3}\}$, $(\mathbf{21,8})\{\mathbf{2;3}\}$, $(\mathbf{22,21})\{\mathbf{3,4;1,2,3}\}$, $(\mathbf{22,35})\{\mathbf{3;4;1,2}\}$, $(\mathbf{33,5})\{\mathbf{1;2}\}$, $(\mathbf{33,10})\{\mathbf{2;3;1,4}\}$, $(\mathbf{38,7})\{\mathbf{1;3}\}$, $(\mathbf{55,3})\{\mathbf{1,3;2,3}\}$, $(\mathbf{94,1})\{\mathbf{2}\}$, $(\mathbf{95,3})\{\mathbf{2;3}\}$, $(\mathbf{142,1})\{\mathbf{1}\}$, $(\mathbf{142,3})\{\mathbf{1;2,3}\}$, $(\mathbf{161,1})\{\mathbf{1,2}\}$, $(\mathbf{390,1})\{\mathbf{2;4}\}$, $(\mathbf{390,7})\{\mathbf{2;5;1,3,4}\}$, $(\mathbf{566,1})\{\mathbf{1;2}\}$, $(\mathbf{770,1})\{\mathbf{1,3;2,4}\}$, $(\mathbf{798,1})\{\mathbf{1;3;4}\}$, $(\mathbf{858,1})\{\mathbf{2;3;4}\}$, $(\mathbf{910,1})\{\mathbf{1;3;4}\}$, $(\mathbf{930,1})\{\mathbf{2;3;4}\}$, $(\mathbf{966,1})\{\mathbf{2;1,3;3,4}\}$, $(\mathbf{1230,1})\{\mathbf{1;2;4}\}$, $(\mathbf{1230,1})\{\mathbf{1;2,3;3,4}\}$, $(\mathbf{1290,1})\{\mathbf{3;1,2;2,4}\}$, $(\mathbf{1410,1})\{\mathbf{4;1,3;2,3}\}$, $(\mathbf{1482,1})\{\mathbf{1;4;2,3}\}$, $(\mathbf{1482,1})\{\mathbf{1;3;2,4}\}$
2	17^3	$x^2+280x+29392$	$(462,5)\{2;3;1,5;4,5\}$, $(570,7)\{4;5;1,3;2,3\}$, $(858,1)\{2;1,3;3,4\}$, $(910,1)\{4;1,3;2,3\}$, $(1230,1)\{2;3;4\}$, $(1290,1)\{1;2,3;3,4\}$
2	19^2	$x^2+76x+2128$	$(6,25)\{1,2\}$, $(6,61)\{1;2\}$, $(6,73)\{1,3;2,3\}$, $(6,77)\{1;2;3\}$, $(6,143)\{1;2,3;3,4\}$, $(6,145)\{3,4;1,2,3\}$, $(6,161)\{4;1,3;2,3\}$, $(6,235)\{2;1,3;3,4\}$, $(6,235)\{3;1,2;2,4\}$, $(10,21)\{2;4\}$, $(10,21)\{4;1,2,3\}$, $(10,77)\{3;1,2,4\}$, $(10,93)\{2;3;1,4\}$, $(14,17)\{1;3\}$, $(14,51)\{1;4;2,3\}$, $(22,13)\{1;2\}$, $(22,21)\{1;3;4\}$, $(26,51)\{1;2;3;4\}$, $(33,4)\{3;1,2\}$, $(33,14)\{2;3;4\}$, $(33,28)\{1;2;3;4\}$, $(34,37)\{1;2;3\}$, $(46,35)\{1;2;3;4\}$, $(55,1)\{2\}$, $(55,2)\{3;1,2\}$, $(55,3)\{1;3\}$, $(55,3)\{3;1,2\}$, $(143,2)\{2;3\}$, $(143,4)\{1;2;3\}$, $(210,11)\{1;4;2,5;3,5\}$, $(210,11)\{2;3;1,5;4,5\}$, $(210,11)\{3;5;1,2;2,4\}$, $(210,11)\{5;1,3;2,4;3,4\}$, $(210,11)\{1,5;2,5;3,5;4,5\}$, $(210,13)\{3;1,5;2,5;4,5\}$, $(210,13)\{5;1,3;2,4;3,4\}$, $(330,7)\{1;5;2,3;3,4\}$, $(330,7)\{2;3;4;5\}$, $(330,7)\{3;4;1,5;2,5\}$, $(390,7)\{3;4;5;1,2\}$, $(390,7)\{2;3;1,5;4,5\}$, $(390,7)\{4;1,5;2,5;3,5\}$, $(398,1)\{1;2\}$, $(462,5)\{1;3;4;5\}$, $(462,5)\{2;5;1,3;3,4\}$, $(546,5)\{2;4;1,5;3,5\}$, $(690,1)\{2,3;1,2,4\}$, $(858,1)\{1;3;4\}$, $(930,1)\{3;4;1,2\}$, $(966,1)\{1;2;4\}$, $(1122,1)\{2;4;1,3\}$, $(1122,1)\{2;3;4\}$, $(1326,1)\{1;3;2,4\}$, $(1410,1)\{4;1,3;2,3\}$, $(1410,1)\{2;3;4\}$

TABLE 5. 251 maximal curves over $\mathbb{F}_q$ of genus $3 \le g \le 4$.

g	q	$h_W(x)$	Curves
3	2	x^3+4x^2+3x-1	$(35,17)\{1;2;3\}$, $(267,1)\{1;2\}$, $(1995,1)\{1;2;3;4\}$
3	2^2	$x^3+9x^2+33x+63$	$(15,7)\{2\}$, $(15,19)\{1,2,3\}$, $(15,31)\{2;1,3\}$, $(15,61)\{2;1,3\}$, $(21,5)\{1\}$, $(21,19)\{1,2,3\}$, $(21,25)\{1;2,3\}$, $(21,31)\{1;2,3\}$, $(33,23)\{2;1,3\}$, $(39,11)\{3;1,2\}$, $(55,7)\{1;2,3\}$, $(65,7)\{2;1,3\}$, $(95,7)\{1;2;3\}$, $(203,3)\{1;2;3\}$, $(497,1)\{1;2\}$, $(1785,1)\{1;3;2,4\}$, $(1785,1)\{3;1,2;2,4\}$, $(2415,1)\{1;2;3,4\}$
3	3	x^3+6x^2+9x	$(205,1)\{1;2\}$
		$x^3+6x^2+10x+4$	$(\mathbf{10,59})\{\mathbf{1;2,3}\}$, $(\mathbf{85,2})\{\mathbf{1;2,3}\}$, $(\mathbf{218,1})\{\mathbf{1,2}\}$
3	3^2	$x^3+18x^2+126x+432$	$(10,161)\{1;3;2,4\}$, $(14,95)\{1;2;3,4\}$, $(35,26)\{4;1,3;2,3\}$, $(46,11)\{3;1,2\}$, $(82,13)\{1;2;3\}$, $(106,13)\{1;2;3\}$, $(215,2)\{1;2,3\}$, $(391,1)\{1;2\}$, $(535,1)\{1;2\}$, $(2210,1)\{2;4;1,3\}$
3	5^2	$x^3+30x^2+360x+2200$	$(51,23)\{1;2;3\}$, $(111,4)\{1,3;2,3\}$, $(497,1)\{1;2\}$, $(2442,1)\{2;4;1,3\}$
3	5^3	$x^3+66x^2+1812x+26488$	$(6,143)\{4;1,3\}$, $(14,43)\{1;3\}$, $(39,2)\{3\}$, $(39,4)\{1;3\}$, $(183,1)\{2\}$, $(1122,1)\{1,3;1,4\}$, $(1218,1)\{3;1,4\}$
3	7	$x^3+12x^2+48x+64$	$(6,55)\{1,4;2,3\}$, $(6,95)\{2;1,3,4\}$, $(10,9)\{1,3\}$, $(10,39)\{1,4;2,3\}$, $(65,2)\{1,3;2,3\}$

3	7^2	$x^3+42x^2+714x+6272$	$(6,59)\{1,3\}$, $(6,85)\{1;2,3\}$, $(6,97)\{1;2\}$, $(6,115)\{1,3;1,4\}$, $(10,39)\{2;1,3\}$, $(10,61)\{1;3\}$, $(10,69)\{3;1,4\}$, $(15,4)\{3\}$, $(15,16)\{2;3\}$, $(15,16)\{3;1,2\}$, $(15,22)\{3;4\}$, $(15,34)\{1,4;2,3,4\}$, $(15,41)\{3;1,2\}$, $(15,44)\{1;3;4\}$, $(15,44)\{1,3;2,4;3,4\}$, $(26,17)\{1;2\}$, $(51,4)\{2;3\}$, $(51,4)\{3;1,2\}$, $(51,10)\{4;1,3;2,3\}$, $(85,11)\{1;2;3\}$, $(226,1)\{2\}$, $(330,1)\{1,3\}$, $(690,1)\{1;2\}$, $(690,1)\{2;3\}$, $(690,1)\{2;4\}$, $(690,1)\{2;1,3\}$, $(858,1)\{3;1,4\}$
3	11^2	$x^3+66x^2+1782x+25168$	$(6,35)\{2;3\}$, $(6,65)\{2;3,4\}$, $(6,83)\{2;3\}$, $(6,85)\{4;1,3;2,3\}$, $(6,95)\{2,3;1,2,4\}$, $(6,107)\{2;3\}$, $(6,161)\{1;2;4\}$, $(6,161)\{2;3;4\}$, $(10,9)\{1,3\}$, $(10,39)\{1,3;2,4\}$, $(10,39)\{1,3;2,3,4\}$, $(10,51)\{2;3;4\}$, $(10,57)\{1;2;3\}$, $(10,57)\{4;1,3;2,3\}$, $(14,15)\{3;4\}$, $(14,15)\{4;2,3\}$, $(14,57)\{2,3;1,2,4\}$, $(15,8)\{1;2\}$, $(15,14)\{2;1,3\}$, $(15,14)\{2;1,4\}$, $(26,15)\{3;1,4\}$, $(39,8)\{2;3\}$, $(39,16)\{1;2;3\}$, $(86,3)\{2;3\}$, $(95,4)\{1,3;2,3\}$, $(210,13)\{2;4;1,3,5\}$, $(210,13)\{1;2;4;5\}$, $(390,7)\{2;3;4;5\}$, $(510,1)\{3;1,2\}$, $(510,1)\{2,4;1,2,3\}$, $(510,7)\{3;4;1,5;2,5\}$, $(510,7)\{1;2,5;3,5;4,5\}$, $(870,1)\{3;4\}$, $(966,1)\{1;2;3\}$, $(1290,1)\{1;2;4\}$, $(1794,1)\{1,3;2,4;3,4\}$
3	11^5	$x^3+2406x^2+2412732x+1290774088$	$(10,119)\{2;3;1,4\}$, $(21,41)\{1;2;3\}$, $(21,73)\{1;2;3\}$, $(39,7)\{1;2\}$, $(86,19)\{1;2;3\}$, $(798,1)\{2,4;1,2,3\}$, $(1974,1)\{3;4;1,2\}$, $(2190,1)\{2;3;1,4\}$, $(2190,1)\{3;4;1,2\}$
3	13	$x^3+18x^2+108x+216$	$(3102,1)\{1;2;3;4\}$
3	13^2	$x^3+78x^2+2496x+41912$	$(6,145)\{1;2;3\}$, $(6,145)\{3;1,2;2,4\}$, $(15,41)\{3;1,2\}$, $(57,10)\{1;4;2,3\}$, $(510,7)\{2;3;4;5\}$, $(1190,1)\{2;1,3;3,4\}$, $(1230,1)\{1;2;3\}$, $(1254,1)\{1;3;4\}$, $(1938,1)\{3;4;1,2\}$, $(1974,1)\{3;4;1,2\}$, $(2910,1)\{3;4;1,2\}$
3	17^2	$x^3+102x^2+4284x+94792$	$(6,107)\{2;3\}$, $(6,161)\{2;3;4\}$, $(10,53)\{2;3\}$, $(10,91)\{1;2,3;3,4\}$, $(14,55)\{1,3;2,4;3,4\}$, $(330,13)\{1;4;2,5;3,5\}$, $(910,1)\{1;3,4\}$, $(966,1)\{1,4;2,3,4\}$, $(1290,1)\{2;3;4\}$
3	17^3	$x^3+420x^2+73488x+6856640$	$(1590,1)\{1;2;4\}$
3	19^2	$x^3+114x^2+5358x+132848$	$(6,221)\{1;4;2,3\}$, $(10,41)\{1;2\}$, $(26,11)\{1;3\}$, $(33,2)\{3\}$, $(33,4)\{1;3\}$, $(158,1)\{1\}$, $(158,3)\{1;2,3\}$, $(210,11)\{5;1,4;2,3,4\}$, $(210,11)\{2,5;1,3,5;3,4,5\}$, $(210,11)\{1;2;3;4\}$, $(330,7)\{1;2;3;5\}$, $(462,5)\{1;2;4;5\}$, $(690,1)\{4;2,3\}$, $(770,3)\{1;3;4;5\}$, $(770,3)\{3;5;1,2;2,4\}$, $(930,1)\{4;1,2,3\}$, $(1326,1)\{3;1,2,4\}$, $(2670,1)\{2;3;1,4\}$
4	2	$x^4+5x^3+6x^2-2x-4$	$(15,23)\{2;1,3\}$, $(21,17)\{1;2,3\}$
4	2^2	$x^4+10x^3+41x^2+96x+172$	$(15,49)\{2;1,3\}$, $(15,77)\{2;4;1,3\}$, $(21,37)\{1;2,3\}$, $(21,55)\{1;2;3,4\}$, $(33,35)\{2;4;1,3\}$, $(35,19)\{2;1,3\}$
		$x^4+10x^3+41x^2+100x+188$	$(15,77)\{1;4;2,3\}$, $(21,37)\{2;3\}$
		$x^4+10x^3+43x^2+110x+205$	$(15,73)\{3;1,2\}$, $(87,5)\{1;2,3\}$, $(111,5)\{2;1,3\}$, $(287,1)\{1,2\}$, $(299,1)\{1,2\}$, $(321,1)\{1,2\}$, $(3795,1)\{1;2;3;4\}$
4	2^7	$x^4+86x^3+3275x^2+72154x+1007077$	$(287,1)\{1,2\}$, $(321,1)\{1,2\}$, $(3795,1)\{1;2;3;4\}$
4	3^2	$x^4+20x^3+174x^2+860x+2713$	$(26,7)\{3\}$, $(26,55)\{2;4;1,3\}$, $(35,4)\{3\}$, $(35,19)\{2;1,3\}$, $(38,13)\{1,2,3\}$, $(58,13)\{1;2,3\}$, $(62,13)\{2;1,3\}$, $(635,1)\{1;2\}$
4	5^2	$x^4+40x^3+680x^2+6400x+36800$	$(6,77)\{2;4\}$, $(14,9)\{3\}$, $(14,13)\{2\}$, $(14,27)\{2;3\}$, $(14,39)\{3;1,2,4\}$, $(38,3)\{3\}$, $(38,9)\{1;3\}$, $(38,9)\{2;3\}$, $(38,9)\{3;1,2\}$, $(51,13)\{2;1,3\}$, $(122,7)\{3;1,2\}$
4	7^2	$x^4+52x^3+1176x^2+15176x+123872$	$(6,85)\{1;2\}$, $(10,69)\{1;3,4\}$, $(15,26)\{1,4;2,3,4\}$, $(15,34)\{3;2,4\}$, $(15,68)\{1;3;2,4\}$, $(15,82)\{1;4;2,3\}$, $(26,15)\{1;4\}$, $(26,15)\{2;1,4\}$, $(34,13)\{1;3\}$, $(51,13)\{3;1,2\}$, $(194,3)\{1;3\}$, $(390,19)\{1,5;2,5;3,5;4,5\}$, $(690,1)\{1;2,3\}$
		$x^4+52x^3+1182x^2+15340x+125497$	$(22,115)\{1;2;3;4\}$, $(26,61)\{1;2;3\}$, $(38,39)\{2;3;1,4\}$, $(38,39)\{4;1,3;2,3\}$, $(38,43)\{1;3;2\}$, $(74,15)\{1,3;2,4;3,4\}$, $(1482,1)\{1;2;4\}$, $(5106,1)\{1;2;3;4\}$
4	11^2	$x^4+88x^3+3344x^2+71632x+950576$	$(6,65)\{2;3\}$, $(6,85)\{2;4\}$, $(6,85)\{4;3\}$, $(6,95)\{2;3,4\}$, $(10,51)\{1;4\}$, $(10,51)\{4;1,2\}$, $(14,9)\{3\}$, $(14,15)\{4;1,3\}$, $(14,27)\{2;3\}$, $(14,45)\{1;3;4\}$, $(14,45)\{2;3;4\}$, $(14,45)\{2;4;1,3\}$, $(14,45)\{4;1,3;2,3\}$, $(15,14)\{1;2\}$, $(21,10)\{2;4\}$, $(21,20)\{1;2;4\}$, $(119,4)\{2;3\}$, $(966,1)\{1,3;1,4\}$
4	11^5	$x^4+3208x^3+4503384x^2+$ $+3613247392x+1812300626896$	$(21,82)\{1;2;3;4\}$, $(39,2)\{2\}$, $(39,4)\{1;2\}$
4	13^2	$x^4+104x^3+4680x^2+118976x+1876576$	$(58,15)\{1;3;4\}$, $(1914,1)\{4;1,2,3\}$
4	17^2	$x^4+136x^3+8024x^2+268192x+5571920$	$(14,57)\{1;2;3\}$, $(21,44)\{2;3;4\}$, $(1590,1)\{1;2;3\}$,

			$(1590,1)\{3;1,2;2,4\}$
4	19^2	$x^4+152x^3+10032x^2+375440x+8739088$	$(210,11)\{1,5;2,3;4,5\}$

TABLE 6. 63 maximal curves over $\mathbb{F}_q$ of genus $g \geq 5$.

g	q	$h_W(x)$	Curves
5	2^2	$x^5+12x^4+62x^3+192x^2+436x+864$	$(15,149)\{1;3;2\}$, $(21,55)\{1;3;2,4\}$, $(51,11)\{1;2,3\}$, $(185,7)\{1;3;2\}$
5	7^2	$x^5+70x^4+2170x^3+39200x^2+458150x+3665984$	$(6,115)\{1;4\}$, $(10,61)\{1,3\}$, $(15,8)\{3\}$, $(15,16)\{1;3\}$, $(15,41)\{1,2,3\}$, $(51,2)\{3\}$, $(51,4)\{1;3\}$, $(690,1)\{2\}$
5	11^2	$x^5+110x^4+5390x^3+154880x^2+2895530x+37097632$	$(6,85)\{4;1,2\}$, $(6,85)\{4;1,3\}$, $(6,85)\{4;2,3\}$, $(6,167)\{1;3\}$, $(10,51)\{2;4\}$, $(10,51)\{3;4\}$, $(10,51)\{4;2,3\}$, $(210,13)\{2;1,5;4,5\}$
5	11^5	$x^5+4010x^4+7237240x^3+7741577680x^2+$ $+5435351766080x+2617242553904032$	$(4326,1)\{1;3;2,4\}$
5	13^2	$x^5+130x^4+7540x^3+256880x^2+5702060x+86755136$	$(6,323)\{1,3;2,4;3,4\}$
5	17^2	$x^5+170x^4+12920x^3+578000x^2+16877600x+337857184$	$(546,11)\{3;4;5;1,2\}$
5	19^2	$x^5+190x^4+16150x^3+808640x^2+26443250x+592837088$	$(6,235)\{1,4;2,3,4\}$, $(210,11)\{1;4;2,3\}$, $(210,11)\{3;1,2;2,4\}$, $(330,7)\{2;3;5\}$, $(462,5)\{1;4;5\}$, $(770,3)\{1,5;3,5;4,5\}$
6	2^2	$x^6+15x^5+102x^4+425x^3+1254x^2+2925x+6021$	$(161,3)\{1;2,3\}$, $(341,1)\{1,2\}$
6	11^5	$x^6+4812x^5+10614300x^4+14191614560x^3+12809583966480x^2+$ $+8223105443361792x+3849697906629612544$	$(570,13)\{3;1,5;2,5;4,5\}$
6	13^2	$x^6+156x^5+11076x^4+473200x^3+13559208x^2+$ $+275345616x+4096157104$	$(6,145)\{3;1,2\}$
6	19^2	$x^6+228x^5+23712x^4+1487320x^3+62701368x^2+$ $+1875470088x+41036244688$	$(51,13)\{1,3;2,3\}$, $(210,11)\{1,3,5;2,4,5;3,4,5\}$, $(966,5)\{1;2;3;5\}$
7	2	$x^7+7x^6+14x^5-2x^4-31x^3-21x^2+3x+1$	$(\mathbf{1995},\mathbf{1})\{\mathbf{1};\mathbf{2};\mathbf{3},\mathbf{4}\}$, $(\mathbf{1995},\mathbf{1})\{\mathbf{1};\mathbf{3};\mathbf{2},\mathbf{4}\}$
7	2^2	$x^7+16x^6+116x^5+520x^4+1685x^3+4424x^2+10310x+22880$	$(15,161)\{1;3;2,4\}$, $(21,85)\{1,3;2,4;3,4\}$, $(35,33)\{3;4;1,2\}$, $(51,23)\{1;2,3\}$, $(141,5)\{2;1,3\}$
7	7^2	$x^7+98x^6+4410x^5+120736x^4+2248022x^3+30194976x^2+$ $+303918580x+2367501248$	$(15,32)\{1;3\}$, $(22,43)\{1;3\}$, $(51,10)\{4;2,3\}$, $(51,20)\{1;4;2,3\}$
7	11^2	$x^7+154x^6+10934x^5+474320x^4+14051730x^3+301274512x^2+$ $+4843013868x+59903475808$	$(6,161)\{2;4\}$, $(10,57)\{1,3;2,3\}$, $(14,115)\{2;4;1,3\}$, $(390,7)\{2;3,5;4,5\}$, $(510,7)\{3,4;2,5;1,3,5\}$
7	19^2	$x^7+266x^6+32718x^5+2466352x^4+127335530x^3+4768267056x^2+$ $+133902303388x+2881195051808$	$(210,11)\{1,3;2,4;3,4\}$, $(770,3)\{3;5;1,4\}$
8	2	$x^8+8x^7+20x^6+6x^5-41x^4-40x^3+15x^2+24x+3$	$(15,53)\{2;1,3\}$
8	17^2	$x^8+272x^7+34544x^6+2718912x^5+148476640x^4+5969177088x^3+$ $+183047217664x^2+4381941960192x+83293534231424$	$(1590,1)\{3;1,2\}$
9	2^2	$x^9+21x^8+207x^7+1289x^6+5781x^5+20283x^4+59389x^3+$ $+153711x^2+370590x+868772$	$(2415,1)\{1;2,3,4\}$
9	7^2	$x^9+126x^8+7434x^7+272832x^6+6981030x^5+132370560x^4+$ $+1932021588x^3+22299105024x^2+208173456414x+$ $+1608995153024$	$(15,16)\{3\}$, $(51,4)\{3\}$
9	11^2	$x^9+198x^8+18414x^7+1068672x^6+43379226x^5+$ $+1308681792x^4+30454499196x^3+560623333248x^2+$ $+8321568916914x+101330949451072$	$(6,85)\{4\}$, $(10,51)\{4\}$
10	5^2	$x^{10}+100x^9+4700x^8+138000x^7+2839000x^6+43530000x^5+$ $+517150000x^4+4894500000x^3+37766375000x^2+$	$(38,9)\{3\}$

		$+243135250000x + 1341717650000$	
10	11^2	$x^{10} + 220x^9 + 22880x^8 + 1495560x^7 + 68916760x^6 + 2380818264x^5 +$ $+64039521040x^4 + 1375875676560x^3 + 24058256936120x^2 +$ $+347686130243320x + 4214462420951024$	$(14, 45)\{4; 1, 3\}$

8. Further tables

Table 7. The 12 pairs (D, N) in the set $\mathcal{S}$ for which we do not prove in Part (2) of Corollary 5.4 whether $\mathrm{Aut}(X_0^D(N))$ consists only of Atkin–Lehner involutions.

(D, N)	$g(X_0^D(N))$	(D, N)	$g(X_0^D(N))$	(D, N)	$g(X_0^D(N))$	(D, N)	$g(X_0^D(N))$
$(21, 29)$	29	$(21, 145)$	177	$(69, 17)$	65	$(77, 17)$	89
$(129, 5)$	41	$(177, 5)$	57	$(205, 1)$	13	$(437, 1)$	33
$(497, 1)$	35	$(517, 1)$	37	$(537, 5)$	177	$(989, 1)$	77

Table 8. The 1321 pairs (D, N) in the set $\mathcal{S}$ for which we do not prove in Part (1) of Corollary 5.4 whether $\mathrm{Aut}(X_0^D(N)/W)$ consists only of Atkin–Lehner involutions for all $W \leq W_0(D, N)$.

$(6, 283)$	$(6, 293)$	$(6, 311)$	$(6, 337)$	$(6, 353)$	$(6, 359)$	$(6, 365)$	$(6, 379)$
$(6, 383)$	$(6, 401)$	$(6, 413)$	$(6, 419)$	$(6, 421)$	$(6, 427)$	$(6, 443)$	$(6, 445)$
$(6, 449)$	$(6, 451)$	$(6, 463)$	$(6, 479)$	$(6, 485)$	$(6, 487)$	$(6, 491)$	$(6, 505)$
$(6, 511)$	$(6, 515)$	$(6, 517)$	$(6, 533)$	$(6, 535)$	$(6, 551)$	$(6, 565)$	$(6, 581)$
$(6, 611)$	$(6, 617)$	$(6, 623)$	$(6, 629)$	$(6, 635)$	$(6, 649)$	$(6, 655)$	$(6, 667)$
$(6, 679)$	$(6, 683)$	$(6, 685)$	$(6, 689)$	$(6, 695)$	$(6, 697)$	$(6, 707)$	$(6, 713)$
$(6, 721)$	$(6, 727)$	$(6, 731)$	$(6, 733)$	$(6, 737)$	$(6, 745)$	$(6, 755)$	$(6, 773)$
$(6, 779)$	$(6, 785)$	$(6, 787)$	$(6, 791)$	$(6, 799)$	$(6, 803)$	$(6, 805)$	$(6, 809)$
$(6, 815)$	$(6, 817)$	$(6, 827)$	$(6, 835)$	$(6, 851)$	$(6, 853)$	$(6, 865)$	$(6, 869)$
$(6, 877)$	$(6, 883)$	$(6, 895)$	$(6, 899)$	$(6, 901)$	$(6, 913)$	$(6, 917)$	$(6, 923)$
$(6, 929)$	$(6, 935)$	$(6, 955)$	$(6, 965)$	$(6, 979)$	$(6, 983)$	$(6, 989)$	$(6, 997)$
$(6, 1001)$	$(6, 1007)$	$(6, 1013)$	$(6, 1015)$	$(6, 1019)$	$(6, 1021)$	$(6, 1027)$	$(6, 1031)$
$(6, 1043)$	$(6, 1045)$	$(6, 1049)$	$(6, 1051)$	$(6, 1057)$	$(6, 1063)$	$(6, 1067)$	$(6, 1069)$
$(6, 1073)$	$(6, 1079)$	$(6, 1081)$	$(6, 1085)$	$(6, 1087)$	$(6, 1091)$	$(6, 1093)$	$(6, 1103)$
$(6, 1105)$	$(6, 1109)$	$(6, 1121)$	$(6, 1129)$	$(6, 1135)$	$(6, 1139)$	$(6, 1147)$	$(6, 1157)$
$(6, 1159)$	$(6, 1163)$	$(6, 1165)$	$(6, 1169)$	$(6, 1171)$	$(6, 1201)$	$(6, 1207)$	$(6, 1211)$
$(6, 1223)$	$(6, 1235)$	$(6, 1243)$	$(6, 1255)$	$(6, 1259)$	$(6, 1265)$	$(6, 1273)$	$(6, 1295)$
$(6, 1307)$	$(6, 1309)$	$(6, 1315)$	$(6, 1333)$	$(6, 1343)$	$(6, 1345)$	$(6, 1357)$	$(6, 1379)$
$(6, 1385)$	$(6, 1391)$	$(6, 1397)$	$(6, 1411)$	$(6, 1415)$	$(6, 1435)$	$(6, 1463)$	$(6, 1469)$
$(6, 1505)$	$(6, 1547)$	$(6, 1555)$	$(6, 1561)$	$(6, 1565)$	$(6, 1577)$	$(6, 1591)$	$(6, 1595)$
$(6, 1603)$	$(6, 1615)$	$(6, 1631)$	$(6, 1639)$	$(6, 1643)$	$(6, 1645)$	$(6, 1649)$	$(6, 1661)$
$(10, 181)$	$(10, 187)$	$(10, 191)$	$(10, 197)$	$(10, 201)$	$(10, 211)$	$(10, 213)$	$(10, 217)$
$(10, 219)$	$(10, 223)$	$(10, 229)$	$(10, 237)$	$(10, 249)$	$(10, 253)$	$(10, 263)$	$(10, 267)$
$(10, 271)$	$(10, 277)$	$(10, 287)$	$(10, 291)$	$(10, 299)$	$(10, 301)$	$(10, 303)$	$(10, 309)$
$(10, 319)$	$(10, 321)$	$(10, 327)$	$(10, 329)$	$(10, 339)$	$(10, 357)$	$(10, 377)$	$(10, 381)$
$(10, 391)$	$(10, 393)$	$(10, 399)$	$(10, 411)$	$(10, 417)$	$(10, 427)$	$(10, 429)$	$(10, 447)$

(10, 451)	(10, 473)	(10, 481)	(10, 483)	(10, 491)	(10, 511)	(10, 519)	(10, 533)
(10, 537)	(10, 553)	(10, 561)	(10, 571)	(10, 573)	(10, 579)	(10, 583)	(10, 587)
(10, 593)	(10, 597)	(10, 599)	(10, 609)	(10, 627)	(10, 633)	(10, 651)	(10, 663)
(10, 669)	(10, 671)	(10, 679)	(10, 681)	(10, 703)	(10, 707)	(10, 709)	(10, 717)
(10, 721)	(10, 723)	(10, 737)	(10, 741)	(10, 749)	(10, 753)	(10, 759)	(10, 767)
(10, 777)	(10, 789)	(10, 807)	(10, 817)	(10, 831)	(10, 843)	(10, 861)	(10, 871)
(10, 879)	(10, 897)	(10, 903)	(10, 949)	(10, 951)	(10, 957)	(10, 969)	(10, 987)
(10, 989)	(14, 73)	(14, 83)	(14, 89)	(14, 97)	(14, 123)	(14, 139)	(14, 141)
(14, 143)	(14, 145)	(14, 155)	(14, 157)	(14, 159)	(14, 163)	(14, 177)	(14, 183)
(14, 185)	(14, 195)	(14, 201)	(14, 205)	(14, 213)	(14, 215)	(14, 219)	(14, 221)
(14, 235)	(14, 237)	(14, 249)	(14, 253)	(14, 255)	(14, 263)	(14, 267)	(14, 269)
(14, 285)	(14, 291)	(14, 295)	(14, 305)	(14, 321)	(14, 331)	(14, 341)	(14, 345)
(14, 365)	(14, 373)	(14, 395)	(14, 429)	(14, 435)	(14, 451)	(14, 453)	(14, 465)
(14, 471)	(14, 485)	(14, 489)	(14, 515)	(14, 519)	(14, 527)	(14, 535)	(14, 551)
(14, 555)	(14, 559)	(14, 561)	(14, 579)	(14, 597)	(14, 615)	(14, 645)	(14, 649)
(14, 663)	(14, 705)	(15, 86)	(15, 94)	(15, 97)	(15, 118)	(15, 119)	(15, 122)
(15, 142)	(15, 146)	(15, 149)	(15, 154)	(15, 161)	(15, 166)	(15, 178)	(15, 182)
(15, 194)	(15, 202)	(15, 203)	(15, 238)	(15, 253)	(15, 266)	(15, 278)	(15, 281)
(15, 286)	(15, 298)	(15, 299)	(15, 302)	(15, 314)	(15, 322)	(15, 323)	(15, 326)
(15, 334)	(15, 358)	(15, 362)	(15, 382)	(15, 394)	(15, 406)	(15, 407)	(15, 418)
(15, 442)	(15, 506)	(15, 518)	(15, 574)	(15, 598)	(15, 602)	(15, 638)	(15, 646)
(15, 658)	(21, 29)	(21, 41)	(21, 58)	(21, 62)	(21, 65)	(21, 67)	(21, 74)
(21, 82)	(21, 85)	(21, 86)	(21, 94)	(21, 106)	(21, 110)	(21, 115)	(21, 118)
(21, 130)	(21, 134)	(21, 145)	(21, 146)	(21, 158)	(21, 170)	(21, 178)	(21, 190)
(21, 194)	(21, 206)	(21, 218)	(21, 226)	(21, 230)	(21, 247)	(21, 254)	(21, 262)
(21, 278)	(21, 290)	(21, 370)	(21, 374)	(21, 410)	(21, 418)	(21, 442)	(22, 37)
(22, 53)	(22, 61)	(22, 71)	(22, 83)	(22, 87)	(22, 93)	(22, 95)	(22, 97)
(22, 101)	(22, 103)	(22, 111)	(22, 115)	(22, 123)	(22, 129)	(22, 141)	(22, 145)
(22, 159)	(22, 161)	(22, 183)	(22, 195)	(22, 205)	(22, 215)	(22, 217)	(22, 219)
(22, 233)	(22, 237)	(22, 249)	(22, 255)	(22, 265)	(22, 267)	(22, 273)	(22, 283)
(22, 285)	(22, 287)	(22, 291)	(22, 303)	(22, 305)	(22, 309)	(22, 321)	(22, 327)
(22, 345)	(22, 357)	(22, 377)	(22, 435)	(26, 41)	(26, 53)	(26, 59)	(26, 61)
(26, 69)	(26, 71)	(26, 77)	(26, 79)	(26, 85)	(26, 87)	(26, 89)	(26, 93)
(26, 97)	(26, 103)	(26, 105)	(26, 111)	(26, 115)	(26, 119)	(26, 123)	(26, 127)
(26, 145)	(26, 159)	(26, 165)	(26, 173)	(26, 185)	(26, 205)	(26, 213)	(26, 219)
(26, 231)	(26, 237)	(26, 255)	(26, 265)	(26, 267)	(26, 285)	(26, 295)	(26, 301)
(26, 319)	(26, 345)	(26, 357)	(33, 34)	(33, 46)	(33, 62)	(33, 70)	(33, 74)
(33, 95)	(33, 115)	(33, 118)	(33, 122)	(33, 130)	(33, 142)	(33, 146)	(33, 158)
(33, 190)	(33, 230)	(33, 238)	(33, 266)	(34, 41)	(34, 53)	(34, 55)	(34, 57)
(34, 59)	(34, 65)	(34, 69)	(34, 87)	(34, 91)	(34, 95)	(34, 97)	(34, 105)
(34, 111)	(34, 115)	(34, 123)	(34, 133)	(34, 139)	(34, 155)	(34, 159)	(34, 165)
(34, 183)	(34, 195)	(34, 213)	(34, 215)	(34, 217)	(34, 219)	(34, 231)	(34, 273)
(34, 285)	(35, 31)	(35, 33)	(35, 37)	(35, 38)	(35, 46)	(35, 58)	(35, 59)
(35, 62)	(35, 66)	(35, 74)	(35, 78)	(35, 82)	(35, 102)	(35, 114)	(35, 118)
(35, 138)	(35, 174)	(35, 186)	(38, 29)	(38, 31)	(38, 37)	(38, 43)	(38, 51)
(38, 55)	(38, 67)	(38, 69)	(38, 85)	(38, 87)	(38, 91)	(38, 93)	(38, 97)
(38, 105)	(38, 109)	(38, 119)	(38, 129)	(38, 159)	(38, 165)	(38, 183)	(38, 185)

(38, 195)	(38, 201)	(38, 203)	(38, 205)	(38, 255)	(38, 259)	(39, 17)	(39, 31)
(39, 34)	(39, 35)	(39, 38)	(39, 46)	(39, 62)	(39, 70)	(39, 74)	(39, 110)
(39, 118)	(39, 122)	(39, 133)	(39, 134)	(39, 170)	(39, 190)	(46, 17)	(46, 29)
(46, 31)	(46, 33)	(46, 37)	(46, 39)	(46, 51)	(46, 53)	(46, 55)	(46, 57)
(46, 59)	(46, 65)	(46, 77)	(46, 85)	(46, 87)	(46, 93)	(46, 105)	(46, 111)
(46, 129)	(46, 165)	(46, 195)	(51, 22)	(51, 23)	(51, 29)	(51, 35)	(51, 46)
(51, 47)	(51, 58)	(51, 70)	(51, 94)	(51, 95)	(51, 106)	(51, 110)	(51, 130)
(51, 154)	(55, 21)	(55, 26)	(55, 34)	(55, 42)	(55, 78)	(55, 102)	(55, 114)
(57, 26)	(57, 55)	(57, 58)	(57, 70)	(57, 74)	(57, 82)	(57, 85)	(57, 91)
(57, 110)	(58, 19)	(58, 23)	(58, 33)	(58, 35)	(58, 39)	(58, 51)	(58, 55)
(58, 57)	(58, 65)	(58, 69)	(58, 77)	(58, 89)	(58, 93)	(58, 105)	(58, 123)
(58, 133)	(58, 143)	(58, 165)	(62, 17)	(62, 19)	(62, 23)	(62, 29)	(62, 33)
(62, 35)	(62, 39)	(62, 43)	(62, 51)	(62, 57)	(62, 69)	(62, 77)	(62, 85)
(62, 87)	(62, 105)	(62, 111)	(65, 21)	(65, 22)	(65, 42)	(65, 66)	(65, 102)
(69, 11)	(69, 14)	(69, 17)	(69, 22)	(69, 26)	(69, 34)	(69, 35)	(69, 38)
(69, 55)	(69, 58)	(69, 70)	(69, 110)	(74, 11)	(74, 17)	(74, 19)	(74, 21)
(74, 23)	(74, 33)	(74, 39)	(74, 51)	(74, 65)	(74, 69)	(74, 73)	(74, 87)
(74, 93)	(74, 95)	(74, 105)	(77, 5)	(77, 10)	(77, 15)	(77, 17)	(77, 26)
(77, 30)	(77, 34)	(82, 13)	(82, 17)	(82, 21)	(82, 29)	(82, 33)	(82, 35)
(82, 39)	(82, 51)	(82, 55)	(82, 65)	(82, 77)	(82, 95)	(82, 105)	(85, 14)
(85, 22)	(85, 42)	(85, 66)	(86, 11)	(86, 13)	(86, 15)	(86, 19)	(86, 21)
(86, 31)	(86, 33)	(86, 35)	(86, 39)	(86, 55)	(86, 57)	(86, 69)	(86, 85)
(86, 91)	(86, 95)	(86, 105)	(87, 10)	(87, 14)	(87, 17)	(87, 22)	(87, 34)
(87, 35)	(87, 46)	(87, 70)	(91, 10)	(91, 15)	(91, 30)	(93, 13)	(93, 14)
(93, 22)	(93, 26)	(94, 15)	(94, 21)	(94, 33)	(94, 35)	(94, 57)	(94, 105)
(95, 7)	(95, 14)	(95, 42)	(106, 7)	(106, 11)	(106, 13)	(106, 15)	(106, 17)
(106, 21)	(106, 23)	(106, 33)	(106, 39)	(106, 51)	(106, 55)	(106, 57)	(106, 65)
(111, 10)	(111, 14)	(111, 22)	(111, 26)	(111, 38)	(115, 14)	(115, 42)	(118, 13)
(118, 15)	(118, 17)	(118, 21)	(118, 23)	(118, 33)	(118, 35)	(118, 57)	(119, 11)
(119, 22)	(119, 30)	(122, 11)	(122, 13)	(122, 15)	(122, 21)	(122, 31)	(122, 33)
(122, 35)	(122, 55)	(123, 7)	(123, 10)	(123, 14)	(129, 5)	(129, 7)	(129, 14)
(129, 22)	(129, 26)	(133, 3)	(133, 5)	(133, 6)	(133, 10)	(133, 30)	(134, 7)
(134, 15)	(134, 19)	(134, 21)	(141, 5)	(141, 10)	(141, 14)	(141, 17)	(141, 34)
(142, 7)	(142, 11)	(142, 13)	(142, 15)	(142, 21)	(142, 35)	(142, 39)	(142, 51)
(143, 5)	(143, 6)	(143, 10)	(143, 14)	(143, 30)	(145, 6)	(145, 14)	(146, 11)
(146, 15)	(146, 21)	(146, 31)	(146, 33)	(146, 35)	(155, 6)	(155, 7)	(155, 14)
(158, 5)	(158, 7)	(158, 11)	(158, 13)	(158, 15)	(158, 21)	(158, 31)	(158, 33)
(158, 35)	(158, 39)	(159, 10)	(159, 14)	(161, 6)	(161, 10)	(161, 30)	(166, 5)
(166, 7)	(166, 15)	(166, 21)	(166, 33)	(177, 5)	(177, 10)	(177, 14)	(177, 22)
(177, 26)	(178, 5)	(178, 7)	(178, 13)	(178, 15)	(178, 21)	(178, 33)	(183, 5)
(183, 10)	(183, 22)	(185, 2)	(185, 3)	(185, 6)	(185, 7)	(185, 14)	(187, 3)
(187, 6)	(187, 7)	(187, 10)	(187, 14)	(194, 11)	(194, 13)	(194, 15)	(194, 17)
(194, 19)	(194, 21)	(194, 23)	(194, 33)	(201, 14)	(202, 5)	(202, 11)	(202, 15)
(202, 19)	(202, 21)	(203, 3)	(203, 6)	(203, 10)	(205, 1)	(205, 2)	(205, 3)
(205, 14)	(206, 11)	(206, 13)	(206, 15)	(206, 33)	(209, 2)	(209, 6)	(213, 10)
(214, 5)	(214, 7)	(214, 15)	(214, 21)	(215, 3)	(215, 6)	(217, 6)	(217, 10)
(218, 5)	(218, 7)	(218, 15)	(219, 2)	(219, 5)	(219, 10)	(219, 14)	(221, 1)

(221, 2)	(221, 3)	(221, 6)	(226, 5)	(226, 15)	(235, 3)	(235, 6)	(237, 14)
(247, 6)	(249, 2)	(249, 10)	(253, 6)	(254, 13)	(254, 15)	(254, 21)	(259, 6)
(259, 10)	(262, 3)	(262, 5)	(262, 15)	(265, 2)	(265, 6)	(267, 10)	(267, 14)
(274, 3)	(274, 5)	(274, 15)	(274, 21)	(278, 3)	(278, 7)	(278, 15)	(287, 2)
(287, 3)	(287, 6)	(287, 10)	(291, 2)	(291, 5)	(291, 10)	(291, 14)	(295, 6)
(298, 3)	(298, 5)	(298, 15)	(299, 2)	(299, 6)	(302, 3)	(302, 5)	(302, 15)
(302, 21)	(303, 2)	(303, 10)	(305, 2)	(305, 6)	(309, 2)	(309, 10)	(309, 14)
(314, 3)	(314, 7)	(314, 13)	(314, 21)	(319, 3)	(319, 6)	(321, 2)	(326, 7)
(327, 10)	(329, 2)	(329, 6)	(330, 29)	(334, 3)	(334, 15)	(335, 2)	(339, 2)
(341, 6)	(346, 3)	(346, 5)	(346, 15)	(355, 6)	(358, 3)	(358, 5)	(358, 15)
(362, 3)	(362, 5)	(362, 15)	(365, 2)	(365, 6)	(371, 2)	(371, 6)	(377, 6)
(381, 2)	(382, 3)	(382, 5)	(382, 7)	(382, 15)	(386, 3)	(386, 7)	(386, 15)
(391, 2)	(393, 10)	(394, 3)	(394, 5)	(395, 2)	(395, 6)	(398, 3)	(403, 6)
(407, 6)	(415, 2)	(415, 6)	(417, 2)	(422, 5)	(427, 6)	(437, 1)	(437, 2)
(446, 5)	(446, 11)	(446, 15)	(447, 2)	(453, 2)	(454, 3)	(458, 3)	(458, 5)
(469, 2)	(469, 6)	(471, 2)	(482, 3)	(482, 5)	(485, 6)	(489, 7)	(497, 1)
(497, 2)	(497, 6)	(502, 3)	(505, 6)	(510, 19)	(514, 3)	(517, 1)	(517, 2)
(519, 2)	(526, 5)	(537, 1)	(537, 2)	(537, 5)	(542, 1)	(542, 5)	(551, 1)
(551, 2)	(554, 5)	(570, 17)	(579, 2)	(586, 3)	(586, 5)	(591, 1)	(591, 2)
(597, 2)	(635, 1)	(635, 2)	(669, 1)	(669, 2)	(674, 1)	(674, 3)	(689, 1)
(689, 2)	(690, 13)	(706, 1)	(706, 3)	(734, 5)	(745, 2)	(753, 1)	(753, 2)
(766, 5)	(771, 2)	(778, 5)	(791, 3)	(794, 1)	(798, 11)	(849, 2)	(866, 1)
(866, 5)	(870, 11)	(878, 1)	(886, 3)	(905, 2)	(921, 2)	(922, 3)	(933, 2)
(934, 1)	(958, 3)	(974, 1)	(974, 3)	(989, 1)	(993, 2)	(1006, 3)	(1011, 2)
(1018, 3)	(1047, 2)	(1055, 2)	(1057, 2)	(1059, 2)	(1101, 2)	(1114, 3)	(1119, 2)
(1122, 7)	(1126, 3)	(1142, 1)	(1149, 2)	(1167, 2)	(1191, 2)	(1203, 2)	(1214, 3)
(1227, 1)	(1230, 7)	(1238, 1)	(1254, 7)	(1262, 1)	(1262, 3)	(1290, 7)	(1302, 5)
(1317, 2)	(1322, 3)	(1326, 5)	(1366, 1)	(1454, 3)	(1482, 5)	(1518, 5)	(1554, 5)
(1594, 1)	(1610, 3)	(1722, 5)	(1785, 2)	(1794, 5)	(1806, 5)	(1822, 1)	(1838, 1)
(1870, 3)	(1914, 5)	(1938, 5)	(1974, 5)	(1982, 1)	(1994, 1)	(2002, 3)	(2030, 3)
(2090, 3)	(2186, 1)	(2210, 3)	(2415, 2)	(2470, 3)	(2530, 3)	(2590, 3)	(2618, 3)
(2870, 3)	(2990, 1)	(2990, 3)	(3010, 3)	(3045, 1)	(3045, 2)	(3094, 1)	(3094, 3)
(3102, 1)	(3135, 2)	(3190, 3)	(3198, 1)	(3230, 3)	(3290, 3)	(3315, 2)	(3666, 1)
(3705, 2)	(3770, 1)	(3774, 1)	(3795, 2)	(3910, 1)	(4074, 1)	(4110, 1)	(4130, 1)
(4134, 1)	(4182, 1)	(4270, 1)	(4305, 2)	(4386, 1)	(4422, 1)	(4470, 1)	(4485, 1)
(4485, 2)	(4510, 1)	(4515, 2)	(4674, 1)	(4730, 1)	(4774, 1)	(4785, 2)	(4810, 1)
(4818, 1)	(4845, 1)	(4845, 2)	(4902, 1)	(4935, 2)	(5106, 1)	(5110, 1)	(5187, 1)
(5190, 1)	(5214, 1)	(5330, 1)	(5370, 1)	(5406, 1)	(5478, 1)	(5530, 1)	(5538, 1)
(5730, 1)	(5790, 1)	(5830, 1)	(5874, 1)	(5934, 1)	(6042, 1)	(6105, 1)	(6162, 1)
(6258, 1)	(6279, 1)	(6314, 1)	(6342, 1)	(6402, 1)	(6438, 1)	(6474, 1)	(6486, 1)
(6710, 1)	(6726, 1)	(6790, 1)	(6810, 1)	(6834, 1)	(6882, 1)	(6942, 1)	(6954, 1)
(6990, 1)	(7014, 1)	(7030, 1)	(7070, 1)	(7210, 1)	(7242, 1)	(7266, 1)	(7370, 1)
(7378, 1)	(7458, 1)	(7490, 1)	(7670, 1)	(7714, 1)	(7826, 1)	(7890, 1)	(8058, 1)
(8106, 1)	(8170, 1)	(8274, 1)	(8294, 1)	(8346, 1)	(8382, 1)	(8466, 1)	(8814, 1)
(8890, 1)	(9174, 1)	(9490, 1)	(9546, 1)	(9618, 1)	(9786, 1)	(9842, 1)	(9858, 1)
(9890, 1)							

TABLE 9. The 133 pairs (D, N) for which $X_0^D(N)$ is found to be geometrically tetragonal via the natural degree 2 map to an Atkin–Lehner quotient $X_0^D(N)/\langle w_m \rangle$ of genus $g \leq 2$, as described in the proof of Part (1) of Proposition 6.17.

(D,N)	m	g	(D,N)	m	g	(D,N)	m	g	(D,N)	m	g
$(6,11)$	66	0	$(6,17)$	34	0	$(6,19)$	114	0	$(6,23)$	69	1
$(6,25)$	150	1	$(6,29)$	174	0	$(6,31)$	186	0	$(6,37)$	222	0
$(6,41)$	246	1	$(6,43)$	129	1	$(6,47)$	94	1	$(6,49)$	294	2
$(6,53)$	106	2	$(6,59)$	354	2	$(6,61)$	366	2	$(6,67)$	402	2
$(6,71)$	426	1	$(6,79)$	474	2	$(6,109)$	654	2	$(10,9)$	90	1
$(10,11)$	110	0	$(10,13)$	65	0	$(10,17)$	170	1	$(10,19)$	38	0
$(10,23)$	230	0	$(10,29)$	290	1	$(10,31)$	62	1	$(10,37)$	185	2
$(14,3)$	14	0	$(14,5)$	14	0	$(14,9)$	14	2	$(14,11)$	154	2
$(14,13)$	182	1	$(14,19)$	266	1	$(15,2)$	15	0	$(15,4)$	15	0
$(15,7)$	3	1	$(15,8)$	15	1	$(15,11)$	55	1	$(15,13)$	195	1
$(15,17)$	255	1	$(21,2)$	7	0	$(21,4)$	7	1	$(21,5)$	15	1
$(21,11)$	231	1	$(22,3)$	66	0	$(22,5)$	110	0	$(22,7)$	154	1
$(22,13)$	26	2	$(22,17)$	374	1	$(26,1)$	26	0	$(26,3)$	26	0
$(26,5)$	26	1	$(26,7)$	26	2	$(33,2)$	66	1	$(33,5)$	55	1
$(33,7)$	231	1	$(34,3)$	17	1	$(34,5)$	170	2	$(34,11)$	374	2
$(35,1)$	35	0	$(35,2)$	35	1	$(35,3)$	35	1	$(38,1)$	38	0
$(38,3)$	38	1	$(38,7)$	266	2	$(39,1)$	39	0	$(39,2)$	39	0
$(39,4)$	39	1	$(46,3)$	69	2	$(46,5)$	230	1	$(51,1)$	51	0
$(51,2)$	51	2	$(55,1)$	55	0	$(55,2)$	55	2	$(57,1)$	19	0
$(57,2)$	19	2	$(58,1)$	29	0	$(58,3)$	29	2	$(62,1)$	62	0
$(62,3)$	62	2	$(65,1)$	65	1	$(69,1)$	69	0	$(74,1)$	74	0
$(74,3)$	74	2	$(77,1)$	11	1	$(82,1)$	41	0	$(85,1)$	17	1
$(86,1)$	86	0	$(87,1)$	87	0	$(87,2)$	87	2	$(91,1)$	91	2
$(93,1)$	31	0	$(93,2)$	31	2	$(94,1)$	94	0	$(95,1)$	95	0
$(95,2)$	95	2	$(106,1)$	106	1	$(111,1)$	111	0	$(111,2)$	111	2
$(115,1)$	23	1	$(118,1)$	59	1	$(119,1)$	119	0	$(122,1)$	122	1
$(123,1)$	123	2	$(129,1)$	129	1	$(134,1)$	134	0	$(141,1)$	141	2
$(142,1)$	2	2	$(143,1)$	143	1	$(146,1)$	146	0	$(155,1)$	155	2
$(158,1)$	158	2	$(159,1)$	159	0	$(161,1)$	161	2	$(166,1)$	166	1
$(178,1)$	89	1	$(183,1)$	183	2	$(194,1)$	194	0	$(202,1)$	101	1
$(206,1)$	206	0	$(210,1)$	30	1	$(215,1)$	215	1	$(237,1)$	79	2
$(254,1)$	254	2	$(314,1)$	314	1	$(326,1)$	326	2	$(330,1)$	3	1
$(390,1)$	390	1	$(446,1)$	446	2	$(462,1)$	154	1	$(510,1)$	510	1
$(546,1)$	546	1									

TABLE 10. The 24 pairs (D, N) for which $X_0^D(N)$ is found to be geometrically tetragonal via the natural degree 4 map to an Atkin–Lehner quotient $X_0^D(N)/\langle w_{m_1}, w_{m_2} \rangle$ of genus 0, as described in the proof of Part (1) of Proposition 6.17.

(D,N)	m_1, m_2	(D,N)	m_1, m_2	(D,N)	m_1, m_2
$(6,35)$	$6, 70$	$(6,55)$	$10, 165$	$(6,65)$	$2, 195$

$(6, 77)$	$21, 154$	$(6, 83)$	$6, 249$	$(6, 89)$	$3, 534$
$(6, 91)$	$42, 546$	$(10, 21)$	$3, 35$	$(10, 33)$	$30, 110$
$(10, 53)$	$53, 530$	$(14, 15)$	$30, 105$	$(14, 23)$	$46, 161$
$(21, 13)$	$39, 91$	$(22, 19)$	$38, 209$	$(34, 7)$	$14, 17$
$(39, 5)$	$5, 39$	$(51, 5)$	$5, 255$	$(58, 5)$	$29, 290$
$(82, 3)$	$6, 246$	$(86, 3)$	$6, 129$	$(570, 1)$	$2, 570$
$(690, 1)$	$5, 138$	$(714, 1)$	$3, 238$	$(770, 1)$	$10, 770$

TABLE 11. The 4 pairs (D, N) for which $X_0^D(N)$ is found to be geometrically tetragonal via the existence of a genus 3 quotient $X_0^D(N)/\langle w_{m_1}\rangle$ with a further quotient $X_0^D(N)/\langle w_{m_1}, w_{m_2}\rangle$ of genus 2, as described in the proof of Part (1) of Proposition 6.17.

(D, N)	m_1	m_2	(D, N)	m_1	m_2
$(14, 17)$	238	14	$(21, 8)$	7	3
$(46, 7)$	161	23	$(145, 1)$	145	5

TABLE 12. The 81 pairs (D, N) for which $X_0^D(N)$ is found to be tetragonal over $\mathbb{Q}$ via certain Atkin–Lehner quotients with rational CM points, as described in the proof of Part (2) of Proposition 6.17. For each pair, we provide a Hall divisor $m \parallel DN$ used in the argument, as well as a discriminant Δ of a CM order for which $X_0^D(N)/\langle w_m\rangle$ has a rational point with CM by that order.

(D, N)	m	Δ	(D, N)	m	Δ	(D, N)	m	Δ
$(6, 35)$	210	-19	$(6, 43)$	258	-3	$(6, 47)$	282	-19
$(6, 53)$	318	-4	$(6, 59)$	354	-43	$(6, 61)$	366	-3
$(6, 65)$	390	-4	$(6, 67)$	402	-3	$(6, 79)$	474	-3
$(6, 89)$	534	-4	$(6, 91)$	546	-3	$(6, 109)$	654	-3
$(10, 19)$	95	-8	$(10, 21)$	210	-3	$(10, 31)$	310	-3
$(10, 37)$	370	-3	$(10, 53)$	530	-43	$(14, 11)$	77	-8
$(14, 17)$	238	-4	$(15, 11)$	165	-7	$(22, 13)$	286	-3
$(26, 3)$	39	-8	$(26, 5)$	130	-11	$(26, 7)$	182	-19
$(33, 5)$	165	-4	$(34, 3)$	102	-3	$(34, 5)$	170	-11
$(34, 7)$	238	-3	$(34, 11)$	374	-11	$(35, 3)$	105	-8
$(38, 3)$	114	-11	$(38, 7)$	266	-19	$(39, 5)$	195	-19
$(46, 3)$	69	-8	$(57, 1)$	3	-19	$(57, 2)$	114	-4
$(58, 3)$	174	-3	$(58, 5)$	290	-11	$(62, 3)$	186	-8
$(65, 1)$	65	-7	$(74, 3)$	222	-8	$(77, 1)$	77	-4
$(82, 1)$	82	-3	$(82, 3)$	246	-3	$(85, 1)$	85	-3
$(91, 1)$	91	-91	$(93, 1)$	93	-4	$(93, 2)$	186	-4
$(106, 1)$	53	-8	$(115, 1)$	23	-115	$(118, 1)$	59	-4
$(122, 1)$	61	-8	$(123, 1)$	123	-123	$(129, 1)$	129	-4
$(141, 1)$	141	-3	$(142, 1)$	142	-3	$(143, 1)$	143	-11
$(145, 1)$	145	-3	$(155, 1)$	155	-7	$(158, 1)$	158	-4
$(161, 1)$	161	-4	$(166, 1)$	83	-4	$(178, 1)$	178	-3
$(183, 1)$	183	-7	$(202, 1)$	101	-8	$(210, 1)$	210	-43

$(215,1)$	215	-43	$(237,1)$	237	-4	$(254,1)$	254	-4
$(314,1)$	314	-8	$(326,1)$	326	-4	$(330,1)$	110	-3
$(390,1)$	390	-67	$(446,1)$	446	-4	$(462,1)$	462	-4
$(510,1)$	510	-3	$(546,1)$	546	-67	$(570,1)$	570	-43
$(690,1)$	690	-3	$(714,1)$	714	-163	$(770,1)$	770	-67

TABLE 13. The 322 pairs (D,N) for which $X_0^D(N)$ is found to not be geometrically tetragonal via a Castelnuovo–Severi argument involving an Atkin–Lehner quotient map of degree 2, as described in the proof of Part (3) of Proposition 6.17. We provide an $m \parallel DN$ which suffices for the argument, along with $g = g(X_0^D(N))$ and $g_m = g(X_0^D(N))/\langle w_m \rangle$.

(D,N)	m	g	g_m	(D,N)	m	g	g_m	(D,N)	m	g	g_m
$(6,85)$	34	17	5	$(6,95)$	570	21	7	$(6,97)$	582	15	4
$(6,101)$	606	17	6	$(6,103)$	309	17	6	$(6,107)$	321	19	5
$(6,113)$	678	19	5	$(6,115)$	69	25	9	$(6,119)$	714	25	7
$(6,121)$	726	23	7	$(6,125)$	750	25	8	$(6,127)$	381	21	6
$(6,131)$	786	23	8	$(6,133)$	798	25	9	$(6,137)$	274	23	6
$(6,139)$	834	23	8	$(6,143)$	286	29	9	$(6,145)$	174	29	9
$(6,149)$	894	25	6	$(6,151)$	906	25	6	$(6,155)$	930	33	11
$(6,157)$	157	25	10	$(6,161)$	966	33	11	$(6,163)$	978	27	8
$(6,167)$	334	29	9	$(6,169)$	1014	29	8	$(6,173)$	346	29	10
$(6,179)$	1074	31	8	$(6,181)$	1086	29	8	$(6,191)$	1146	33	9
$(6,193)$	1158	31	10	$(6,197)$	1182	33	12	$(6,199)$	1194	33	10
$(10,39)$	65	17	5	$(10,41)$	410	15	4	$(10,43)$	430	13	4
$(10,47)$	470	17	4	$(10,49)$	98	17	5	$(10,51)$	170	25	7
$(10,57)$	38	25	7	$(10,59)$	590	21	6	$(10,61)$	122	19	5
$(10,63)$	90	33	13	$(10,67)$	670	21	8	$(10,69)$	230	33	7
$(10,71)$	710	25	5	$(10,73)$	365	23	7	$(10,77)$	770	33	9
$(10,79)$	158	25	9	$(10,83)$	830	29	10	$(10,89)$	890	31	10
$(10,91)$	182	33	11	$(10,97)$	485	31	11	$(10,103)$	1030	33	14
$(14,25)$	350	15	4	$(14,27)$	189	19	7	$(14,29)$	406	15	4
$(14,33)$	77	25	9	$(14,37)$	74	19	5	$(14,39)$	546	29	9
$(14,41)$	574	21	7	$(14,43)$	602	23	6	$(14,47)$	329	25	7
$(14,53)$	371	27	10	$(14,59)$	413	31	11	$(14,61)$	854	31	5
$(15,14)$	7	17	5	$(15,19)$	285	13	3	$(15,22)$	55	25	5
$(15,23)$	115	17	5	$(15,26)$	195	29	9	$(15,28)$	7	33	11
$(15,29)$	87	21	5	$(15,31)$	465	21	7	$(15,32)$	15	33	13
$(15,37)$	555	25	9	$(15,41)$	615	29	5	$(15,43)$	645	29	11
$(15,47)$	705	33	11	$(21,10)$	15	17	5	$(21,16)$	336	25	9
$(21,17)$	51	17	5	$(21,19)$	399	21	3	$(21,23)$	483	25	9
$(21,25)$	175	29	9	$(21,29)$	609	29	11	$(21,31)$	651	33	9
$(22,15)$	110	21	5	$(22,21)$	77	25	9	$(22,23)$	506	21	4
$(22,25)$	550	25	10	$(22,27)$	594	31	10	$(22,29)$	638	25	8
$(22,31)$	341	25	6	$(22,37)$	814	29	12	$(26,9)$	234	13	4
$(26,11)$	286	13	4	$(26,15)$	26	25	7	$(26,17)$	221	19	6
$(26,19)$	494	21	4	$(26,21)$	546	33	11	$(26,23)$	299	25	9

(26, 25)	650	31	10	(26, 29)	754	31	11	(26, 31)	806	33	10
(33, 8)	264	21	7	(33, 10)	55	29	7	(33, 13)	429	21	7
(33, 17)	187	29	11	(33, 19)	627	33	13	(34, 9)	306	17	5
(34, 15)	170	33	11	(34, 19)	646	25	9	(34, 23)	782	33	11
(35, 4)	35	13	4	(35, 6)	35	25	7	(35, 8)	7	25	9
(35, 9)	35	25	9	(35, 11)	35	25	9	(35, 13)	455	29	5
(38, 9)	342	19	7	(38, 11)	209	19	5	(38, 13)	494	21	4
(38, 17)	646	27	10	(39, 7)	91	17	5	(39, 8)	39	25	5
(39, 11)	429	25	9	(46, 11)	506	23	5	(46, 13)	299	23	8
(46, 17)	782	33	11	(51, 4)	204	17	6	(51, 8)	136	33	13
(51, 11)	187	33	13	(55, 4)	55	21	5	(55, 7)	55	25	9
(57, 4)	19	19	7	(57, 5)	19	17	5	(57, 7)	399	25	5
(58, 7)	406	17	5	(58, 9)	29	29	12	(58, 11)	638	29	10
(58, 13)	754	31	11	(62, 7)	434	21	5	(62, 9)	62	31	12
(62, 11)	341	31	9	(65, 3)	195	17	5	(65, 4)	260	25	9
(65, 7)	455	33	7	(69, 4)	276	23	8	(69, 5)	69	21	7
(69, 7)	69	29	11	(74, 5)	74	19	5	(74, 7)	518	25	9
(77, 2)	11	15	5	(77, 3)	231	21	5	(77, 4)	308	31	12
(77, 5)	11	29	11	(82, 5)	410	21	7	(82, 7)	574	25	9
(85, 2)	170	17	6	(85, 3)	255	21	5	(85, 4)	68	33	13
(86, 5)	86	21	6	(86, 7)	602	29	9	(87, 4)	87	29	6
(87, 5)	145	29	11	(91, 2)	91	19	7	(91, 3)	39	25	9
(93, 4)	31	31	7	(93, 5)	31	29	9	(94, 5)	470	23	7
(94, 7)	329	29	9	(95, 3)	95	25	5	(106, 3)	53	17	6
(106, 5)	530	27	7	(106, 7)	371	33	13	(115, 2)	23	23	6
(115, 3)	23	29	9	(118, 3)	354	19	6	(118, 5)	590	29	10
(119, 2)	119	25	3	(119, 3)	119	33	7	(122, 3)	122	21	6
(122, 5)	305	31	12	(123, 2)	123	21	8	(129, 2)	129	21	8
(133, 2)	266	27	9	(134, 3)	134	23	5	(134, 5)	134	33	10
(141, 2)	94	23	8	(142, 3)	426	23	6	(143, 2)	143	31	6
(145, 2)	290	29	10	(146, 3)	146	25	5	(155, 2)	155	31	10
(158, 3)	474	27	9	(159, 2)	159	27	4	(161, 2)	23	33	11
(166, 3)	249	27	11	(177, 2)	354	29	11	(178, 3)	89	29	9
(183, 2)	183	31	8	(185, 1)	185	13	3	(194, 3)	194	33	7
(201, 1)	201	11	3	(201, 2)	402	33	13	(202, 3)	101	33	10
(203, 1)	203	15	4	(209, 1)	209	15	3	(218, 1)	218	10	3
(219, 1)	219	13	3	(221, 1)	221	17	5	(235, 1)	47	15	3
(247, 1)	247	19	7	(249, 1)	249	13	4	(253, 1)	23	17	6
(259, 1)	259	19	6	(265, 1)	53	17	6	(274, 1)	274	11	3
(278, 1)	278	12	3	(287, 1)	287	21	4	(291, 1)	291	17	5
(295, 1)	295	19	6	(299, 1)	299	23	4	(302, 1)	302	13	4
(303, 1)	303	17	4	(305, 1)	305	21	7	(309, 1)	103	17	4
(319, 1)	319	23	7	(321, 1)	321	17	4	(323, 1)	323	25	9
(327, 1)	327	19	4	(329, 1)	329	23	6	(334, 1)	334	13	4
(335, 1)	335	23	3	(339, 1)	339	19	4	(341, 1)	341	25	6
(346, 1)	173	14	4	(355, 1)	355	23	8	(362, 1)	362	16	4
(365, 1)	365	25	8	(371, 1)	371	27	6	(377, 1)	377	29	11

$(381,1)$	127	21	6	$(386,1)$	386	17	4	$(391,1)$	391	29	8
$(393,1)$	393	21	8	$(394,1)$	394	16	6	$(395,1)$	395	27	6
$(398,1)$	398	17	4	$(403,1)$	31	31	13	$(407,1)$	407	31	8
$(411,1)$	411	23	6	$(413,1)$	413	29	10	$(415,1)$	83	27	8
$(417,1)$	139	23	6	$(422,1)$	422	18	7	$(437,1)$	437	33	12
$(447,1)$	447	25	6	$(451,1)$	451	33	11	$(453,1)$	151	25	6
$(454,1)$	454	18	6	$(458,1)$	458	20	4	$(466,1)$	233	19	7
$(469,1)$	469	33	13	$(471,1)$	471	27	6	$(482,1)$	482	21	6
$(485,1)$	485	33	12	$(489,1)$	489	27	9	$(501,1)$	501	27	10
$(502,1)$	502	20	7	$(514,1)$	257	21	7	$(519,1)$	519	29	6
$(526,1)$	526	21	8	$(537,1)$	537	29	12	$(538,1)$	269	22	6
$(542,1)$	542	23	6	$(543,1)$	543	31	10	$(554,1)$	554	24	7
$(562,1)$	281	23	7	$(566,1)$	566	24	5	$(573,1)$	573	31	12
$(579,1)$	579	33	9	$(586,1)$	586	24	8	$(591,1)$	591	33	6
$(597,1)$	199	33	8	$(614,1)$	614	26	5	$(622,1)$	622	25	10
$(626,1)$	626	27	5	$(634,1)$	634	26	10	$(662,1)$	662	28	9
$(674,1)$	674	29	9	$(694,1)$	694	28	12	$(698,1)$	698	30	9
$(706,1)$	706	29	9	$(718,1)$	718	29	12	$(734,1)$	734	31	6
$(746,1)$	746	32	10	$(758,1)$	758	32	11	$(766,1)$	766	31	10
$(778,1)$	389	32	11	$(794,1)$	794	34	7	$(798,1)$	114	17	5
$(802,1)$	401	33	12	$(838,1)$	419	34	13	$(858,1)$	286	21	5
$(870,1)$	145	17	5	$(910,1)$	182	25	7	$(930,1)$	930	21	5
$(966,1)$	966	21	5	$(1110,1)$	222	25	7	$(1122,1)$	34	25	9
$(1190,1)$	1190	33	7	$(1218,1)$	406	29	7	$(1230,1)$	1230	25	7
$(1254,1)$	1254	29	9	$(1290,1)$	430	29	9	$(1302,1)$	186	29	9
$(1326,1)$	1326	33	7	$(1410,1)$	1410	29	7	$(1518,1)$	1518	33	11
$(1590,1)$	1590	33	9								

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